

Design and Analysis of Second-Order Active Filters



**Cal Poly
Pomona**

College of Engineering

Sarah Boyd, Nisarg Ahir, Juan Estrada

Dr. Moussavi

ETE 3101 Circuit Analysis II Laboratory

Lab 4

April 8, 2026

TABLE OF CONTENTS

1. ABSTRACT.....	[Page #3]
2. INTRODUCTION	[Page #3]
3. REVIEW.....	[Page #4]
3a. Conceptual analysis.....	[Page #4]
3b. Single order RC passive filter transfer function table.....	[Page #5]
3c. Active first order transfer function table	[Page #6]
3d. Active second order transfer function table	[Page #7]
3e. Dampening ratio analysis table.....	[Page #9]
3f. Analysis of an underdampened system table.....	[Page #10]
4.METHODOLOGY.....	[Page #10]
5. CIRCUITS.....	[Page#11]
5a. Band pass filter: first configuration.....	[Page #12]
5a1. Determining component values.....	[Page #11]
5a2. Simulated circuit and data.....	[Page #13]
5a3. Hardware Implementation.....	[Page #14]
5a4.Calculations, analysis, and results	[Page# 14]
5b. Band pass filter: second configuration	[Page#22]
5b1. Determining component values.....	[Page #22]
5b2. Simulated circuit and data.....	[Page #23]
5b3.Calculations, analysis, and results	[Page#24]
5c. Band pass filter: third configuration.....	[Page #31]
5c1. Determining component values.....	[Page #32]
5c2. Simulated circuit and data.....	[Page #32]
5c3. Hardware Implementation.....	[Page #33]
5c4.Calculations, analysis, and results.....	[Page#33]
5d. Band pass filter: fourth configuration	[Page#40]
5d1. Determining component values.....	[Page #41]
5d2. Simulated circuit and data.....	[Page #41]
5d3.Calculations, analysis, and results	[Page #43]
6. DISCUSSION AND CONCLUSION	[Page #48]
7. REFERENCES.....	[Page #51]

ABSTRACT

Four filter configurations were analyzed in this experiment. The first was a non-inverting band-pass filter using two stages: a high-pass filter followed by a low-pass filter. The second configuration included three stages: a high-pass filter, an amplifier stage for added gain, and a low-pass filter. The third configuration also used three stages (high-pass, op-amp, and low-pass), but the op-amp operated at unity gain, acting as a buffer. The fourth configuration was an inverting band-pass filter.

Transfer functions for all configurations were derived analytically and verified using Multisim simulations. The first and third configurations were also implemented on a breadboard and tested experimentally. Theoretical cutoff frequencies and -3 dB points closely matched simulated and measured results, confirming the expected filter behavior and stability.

INTRODUCTION

Filters are circuits designed to pass signals within a specified frequency range while attenuating unwanted frequency components. They are widely used in signal processing, communication systems, and instrumentation. Filters can be implemented as either passive filters (as explored in Labs 1 and 3) or active filters (as in Lab 2 and this experiment). They are also classified by order, with this report focusing specifically on second-order filters.

This report investigates the design and behavior of second-order active filters using operational amplifiers and passive RC networks. The objective is to demonstrate the functionality of circuits utilizing one or two operational amplifiers, where the op-amp may act as a buffer (unity gain) or provide amplification, in both inverting and non-inverting configurations.

All filters in this experiment were designed as band-pass filters to enable consistent comparison of data and Bode plots. The filters were designed, simulated, and experimentally tested, following the methodology outlined in Chapter 15 of the referenced textbook, as cited in the references section.

REVIEW

Filters can be classified as active or passive. Electronic components can be classified as passive or active devices. Passive devices include resistors, capacitors, inductors, and diodes. These components do not provide a gain in power—they just store, dissipate, or direct energy. Active devices include transistors and op amps (operational amplifiers). These components require an external power source and are capable of providing amplification. When passive elements (such as an RC filter circuit) are combined with active elements (such as an op amp), this results in an active circuit.

Filters can also be classified by their order. A first-order circuit contains resistors and capacitors, *or* inductors, and has one zero and one pole. A second-order circuit contains two first-order circuits- resistance, plus two inductive or capacitive elements. The second order has two poles and two zeros. Similarly, a third-order circuit consists of a second-order circuit plus a first-order circuit; resistance plus three inductive or capacitive elements, and three poles. Increasing the order of a filter produces a steeper attenuation beyond the cutoff frequency, which gets the circuit closer to the ideal goal of reducing unwanted frequency components outside of the desired bandwidth. The greater the order the steeper the attenuation past the cutoff frequency. To analyze filter behavior, the transfer function of the circuit is determined and analyzed for zero (numerator) and poles (denominator).

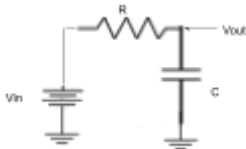
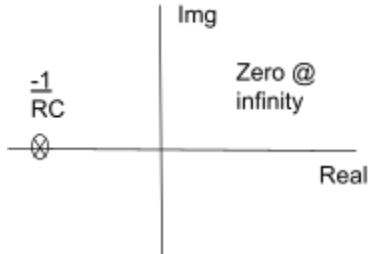
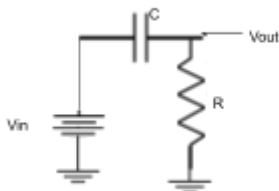
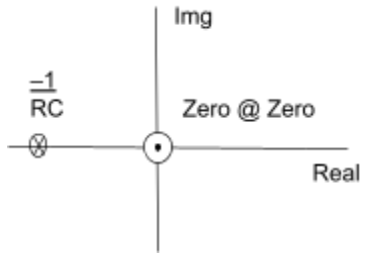
Filter type	f_c	Configuration of passive devices	Transfer function	Stability analysis
Low pass	$f_c = \frac{1}{R\pi 2C}$		$H(S) = \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$	$\frac{\frac{1}{RC}}{S + \frac{1}{RC}} = \frac{n(Z)}{n(P)}$, so $n(Z) = \frac{1}{RC}$ and $n(p) = S + \frac{1}{RC}$. Setting $n(Z) = 0$ gives $\frac{1}{RC} = 0$. Since C cannot be zero, the constant $\frac{1}{RC}$ cannot equal 0. Therefore, we have a zero at infinity and thus a low pass filter. Setting $n(p) = 0$ gives $S + \frac{1}{RC} = 0$ and $S = -\frac{1}{RC}$. We have one pole on the negative side of the s plane, so the system is stable. 
High pass	$f_c = \frac{1}{R\pi 2C}$		$H(S) = \frac{S}{S + \frac{1}{RC}}$	$\frac{S}{S + \frac{1}{RC}} = \frac{n(Z)}{n(P)}$, so $n(Z) = S$ and $n(p) = S + \frac{1}{RC}$. When we set $n(Z) = 0$, $S = 0$. Since $S = 0$, there is one zero at zero, and therefore, we have a high-pass filter. Setting $n(p) = 0$ gives $S + \frac{1}{RC} = 0$ and $S = -\frac{1}{RC}$. We have one pole on the negative side of the s plane, so the system is stable. 

Table 1. Transfer function and analysis for the first order RC passive filters that are used in our second order active circuits. This table can be referenced while finding the transfer function of active filters.

The circuits in table 1 are filter circuits used in the active first and second order circuits. This can be seen when analyzing table 2 and 3.

	Inverting	Non inverting
--	-----------	---------------

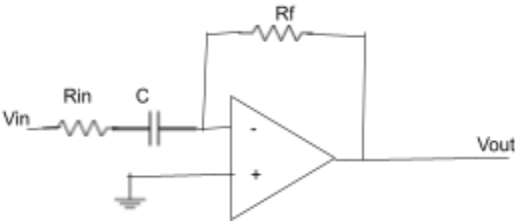
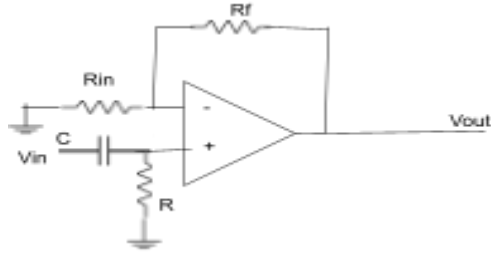
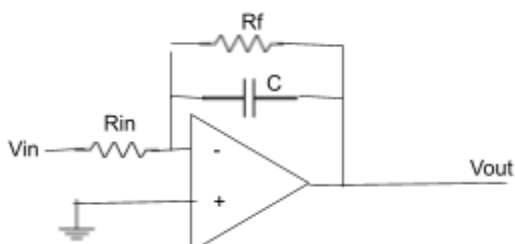
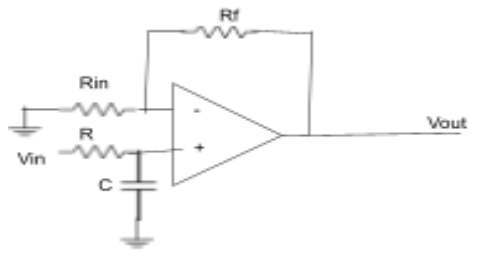
Gain:	$Av = -\frac{R_f}{R_{in}}$	$Av = 1 + \frac{R_f}{R_{in}}$
High pass	 $H(s)_{active} = -\frac{R_f}{R_{in}} \times \frac{S}{S + \frac{1}{R_{in}C}}$	 $H(s)_{active} = \left(1 + \frac{R_f}{R_{in}}\right) \times \frac{S}{S + \frac{1}{RC}}$
Low pass	 $H(s)_{active} = -\frac{R_f}{R_{in}} \times \frac{\frac{1}{RfC}}{S + \frac{1}{RfC}}$	 $H(s)_{active} = \left(1 + \frac{R_f}{R_{in}}\right) \times \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$

Table 2. Transfer functions of active first order filters. In all cases the transfer function is the gain times the transfer function of the passive filter component, or $H(s)_{active} = AV \times H(s)_{passive}$. $H(s)$ is derived from the filter component of the circuit (as seen in table 1)

For a second-order active filter, there are two basic configurations. It can be implemented either as two cascaded first-order active filter stages (i.e., a first-order filter followed by another first-order filter) or by using a buffered op-amp configuration. Each stage can be configured as either a low-pass or a high-pass filter.

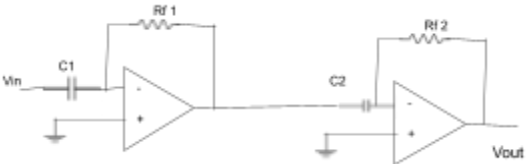
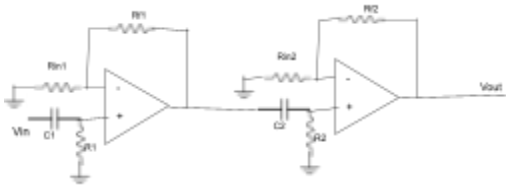
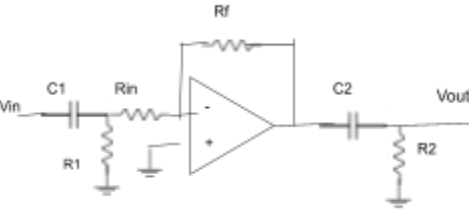
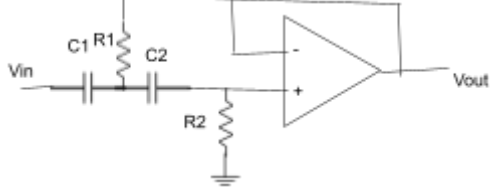
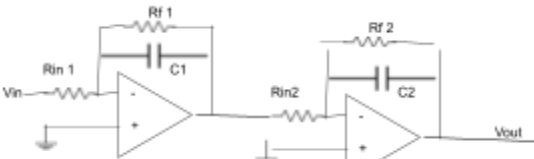
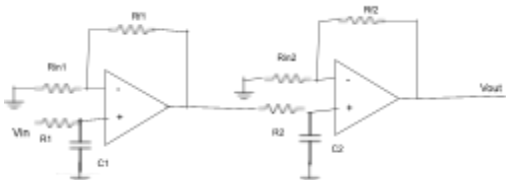
When two high-pass filters are cascaded, the result is a high-pass filter with increased attenuation beyond the cutoff frequency f_c . Similarly, cascading two low-pass filters produces a low-pass filter with greater attenuation beyond f_c . If the first stage is a high-pass filter and the second

stage is a low-pass filter, the overall circuit functions as a band-pass filter. Conversely, if the first stage is a low-pass filter followed by a high-pass filter, the result is a band-stop (notch) filter.

Note that the critical frequency can or can not be equal for each filter. If

$f_{c1} = f_{c2}$, $R1 = R2$, $C1 = C2$. If

$f_{c1} \neq f_{c2}$, $R1 = R2$ and $C1 \neq C2$, or $R1 \neq R2$ and $C1 = C2$, or $R1 \neq R2$ and $C1 \neq C2$.

		Inverting	Non inverting
	Gain:	$Av = -\frac{R_f}{R_{in}}$	$Av = 1 + \frac{R_f}{R_{in}}$
High pass (second order)	Op amps: 2	 $H(s)_{total} = Av_1 \times H(s)_{(stage\ 1)} \times Av_2 \times H(s)_{(stage\ 2)}$	 $H(s)_{total} = Av_1 \times H(s)_{(stage\ 1)} \times Av_2 \times H(s)_{(stage\ 2)}$
	Op amps: 1	 $H(s)_{total} = H(s)_{(stage\ 1)} Av \times H(s)_{(stage\ 2)}$	 Please see the second reference for more information regarding the complete derivation.
Low pass (second order)	Op amps: 2	 $H(s)_{total} = Av_1 \times H(s)_{(stage\ 1)} \times Av_2 \times H(s)_{(stage\ 2)}$	 $H(s)_{total} = Av_1 \times H(s)_{(stage\ 1)} \times Av_2 \times H(s)_{(stage\ 2)}$

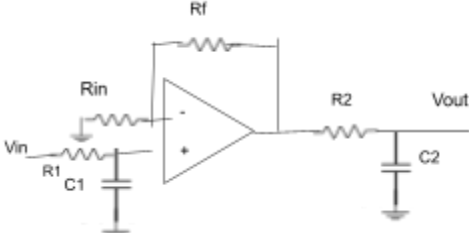
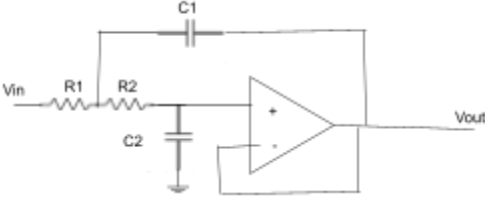
	Op amps: 1	 $H(s)_{total} = H(s)_{(stage\ 1)} A_v \times H(s)_{(stage\ 2)}$	 Please see the third reference for more information regarding the complete derivation.
--	------------	---	---

Table 3. Transfer functions of second order active filters. In all cases the transfer function is the gain times the transfer function of the passive filter component, or $H(s)_{active} = AV \times H(s)_{passive}$. Note that in this table the protective resistors in front of the capacitor were removed. Also note that band pass and band stop were excluded from this table. Note that the one op amp for high pass non inverting is a Sallen Key high pass filter; and that the one op amp for low pass non inverting is a Sallen Key low pass filter.

By knowing the transfer function of stage 1 and stage 2, along with the gain of each stage, we can determine the overall transfer function. The total transfer function is given by: $H(s)_{total} = H(s)_{stage\ 1} \times H(s)_{stage\ 2}$. Next, we substitute appropriate values for R_{in} and R_f based on the desired gain of each stage and simplify the resulting expression. Once simplified, we can extract the key filter characteristics, as summarized in the table below

Stable			
Underdamped		Critically damped	Overdamped
$0 < \xi < 1$		$\xi = 1$	$\xi > 1$
Poles	Two negative, complex poles	Two identical, negative, real poles	Two negative, different, real poles

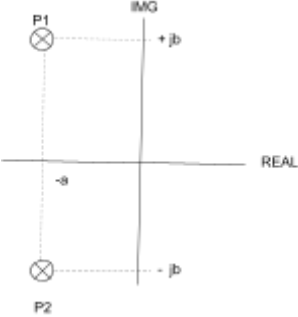
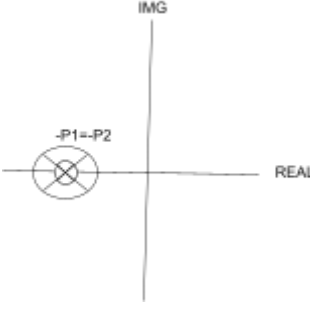
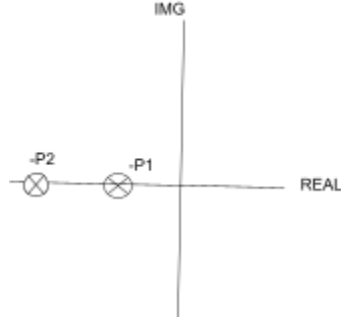
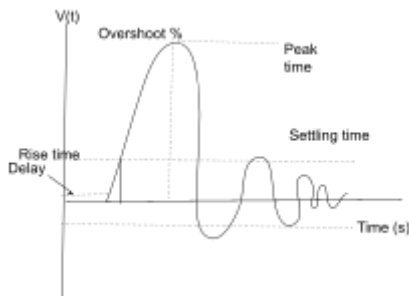
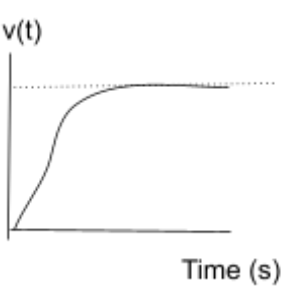
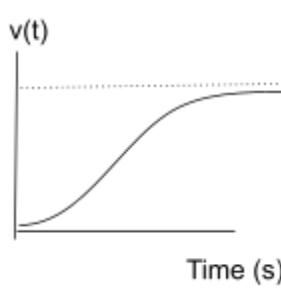
	$P_1 = -a + jb$ $P_2 = -a - jb$		
Stability			
Graph			
Behavior	Oscillating	Fast rise to steady state	Slower rise to steady state
Step Response	Underdamped: $V_o(t) = (1 - e^{\xi\omega_n t}(\cos(\omega_d t) + \frac{\xi}{\beta}\sin(\omega_d t))u(t)$ Critically damped: $V_o(t) = (1 - [e^{-pt} + te^{-pt}])u(t)$ Over damped : $V_o(t) = [1 - (Ae^{-p_1 t} + Be^{-p_2 t})]u(t)$		

Table 3. For critically damped, when doing the quadratic equation to solve for poles the term under the square root must be equivalent to 0.

Regarding the equations for the graph for underdamped,

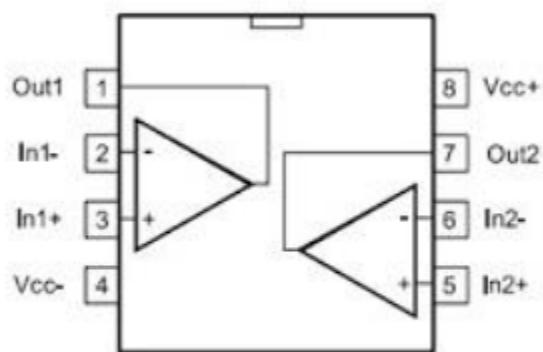
	Formula
Dampening factor	$\beta = \sqrt{1 - \xi^2}$
	$\omega_n = -$

	$\omega_d = \beta \omega_n$
Rise time	$T_R = \frac{1+0.7\xi}{\omega_n}$
Delay time	$T_D = \frac{0.8+2.5\xi}{\omega_n}$
Peak time	$T_p = \frac{\pi}{\omega_d}$
% Overshoot	$O.S\% = e^{\frac{-\pi\xi}{\beta}} * 100$
Settling time	$t_s = \frac{-\ln(\% \text{ in decimal})}{\xi \omega_n}$

Table 4. β and ω_d should be calculated first, in order to find the other formulas. $\omega_o = \omega_n = \omega_r$, but NOT ω_d and ω_c .

METHODOLOGY

All circuits were simulated using Multisim software, and the Bode plot was obtained and analyzed. The first and third configuration were constructed on a bread board. Recall that the first configuration is a non inverting active band pass filter circuit with two operational amplifiers, cascading an active high pass filter with an active low pass filter and that the third configuration consists of three stages- high pass filter, buffer and low pass filter. LM 358p was the op amp selected used for the physical circuits built on the breadboard, and the pinout can be seen below in image 1. Then, the transfer functions were calculated for each circuit and analysis of filter behavior was done.



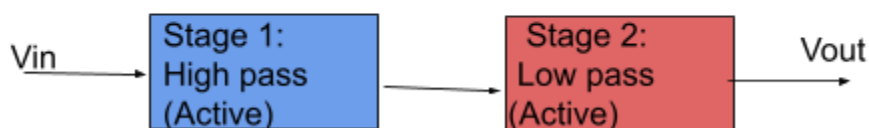
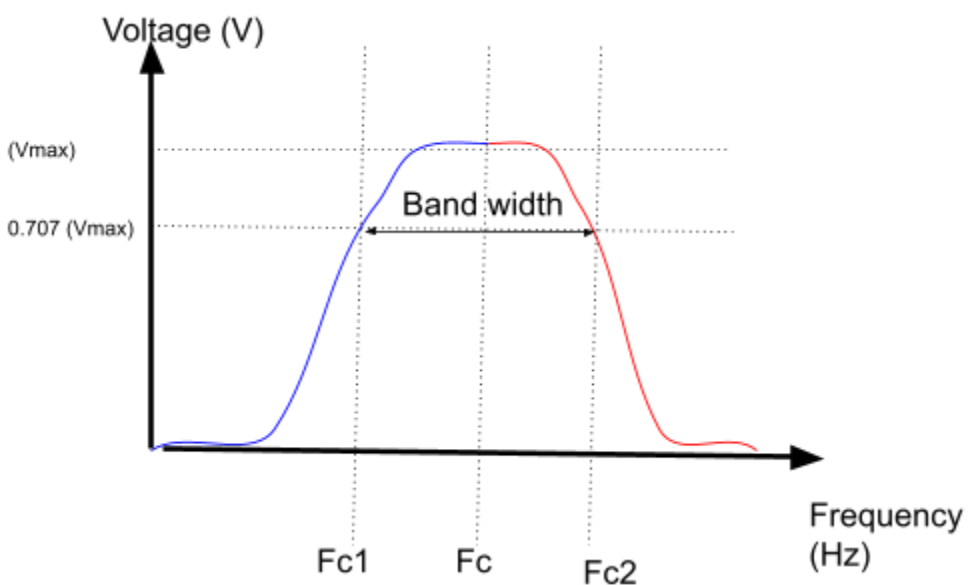
IMG 1. Pinout of the LM358P, from Jotrin.com. This was referenced while building the circuits.

BAND PASS FILTER, FIRST CONFIGURATION: *non inverting active with the first stage being active high pass filter and the second stage being active low pass filter.*

1. Determining component values

$$F_{c1} = 1.59 \text{ kHz and } F_{c2} = 3.18 \text{ kHz}$$

Since $F_{c1} \neq F_{c2}$, and we want the filter to be bandpass, $F_{c1} < F_{c2}$



To increase frequency we decrease R, so $R1=2R$ and $R2=R$. And we can have $C1=C2=C=0.1\mu F$

Check: $F_{c1} = \frac{1}{R1C1} = \frac{1}{2RC}$, so $1.59kHz = \frac{1}{2R\Omega \times 0.1\mu F}$ so $R1 = 2(500\Omega) = 1000\Omega$

Check: $F_{c2} = \frac{1}{R2C2} = \frac{1}{RC}$, so $3.18kHz = \frac{1}{R\Omega \times 0.1\mu F}$ so $R2 = 500\Omega$

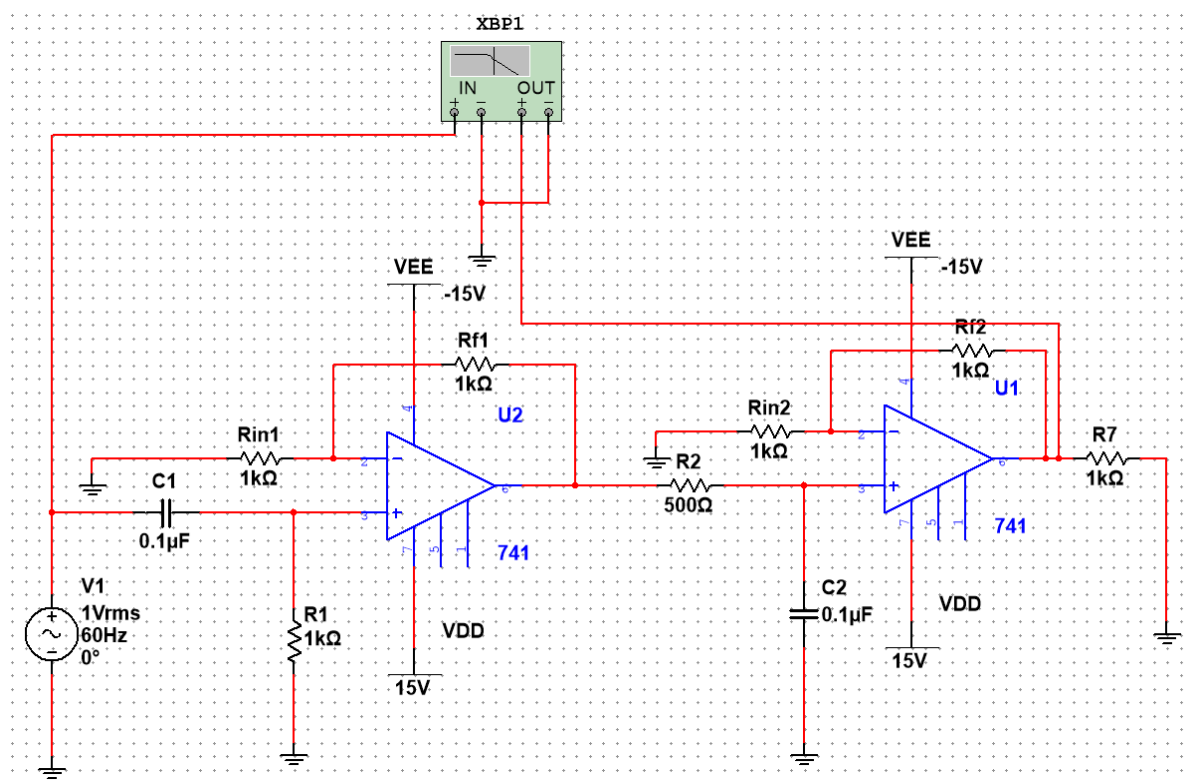
For a gain of 2 for stage 1, $A1 = 1 + \frac{Rf_1}{Rin_1} = 1 + \frac{1k\Omega}{1k\Omega} = 2$

For a gain of 2 for stage 2, $A2 = 1 + \frac{Rf_2}{Rin_2} = 1 + \frac{1k\Omega}{1k\Omega} = 2$

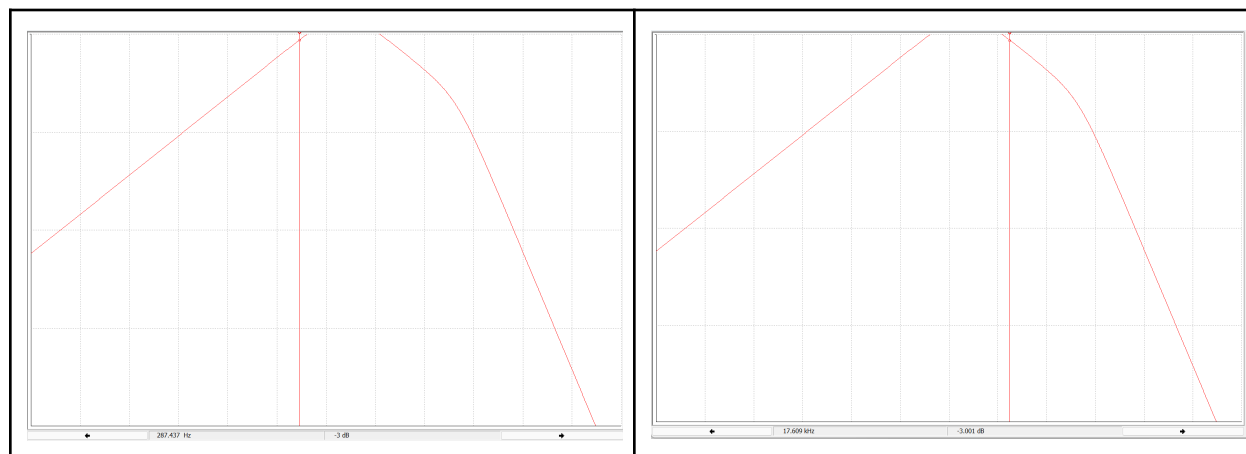
	Circuit from Multisim	Circuit built on breadboard
C=C1=C2	0.1uF	0.1uF
R1	R1=2R=1000	820+180=1k
R2	R2=R=500	470+33=503
Rin_1	1k Ω	1k Ω
Rin_2	1k Ω	1k Ω
Rf_1	1k Ω	1k Ω
Rf_2	1k Ω	1k Ω
Gain	2	2

Table 5. *Component values*

Multisim



IMG 2. Schematic of circuit on Multisim

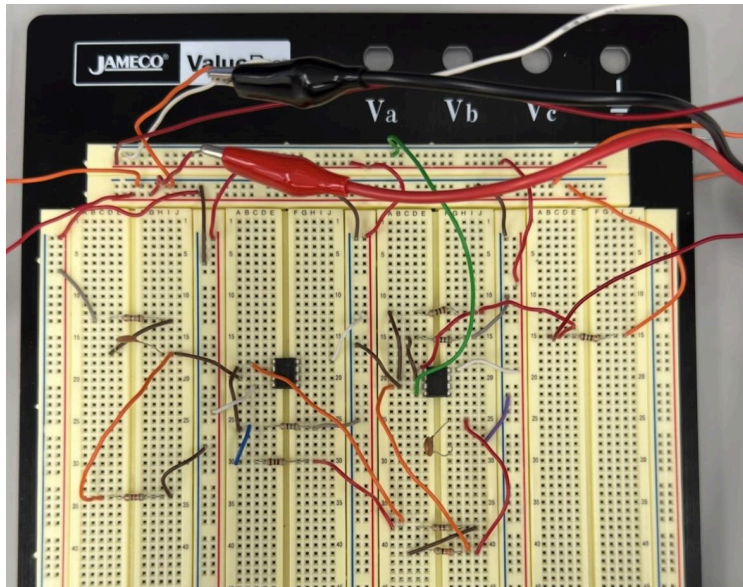


IMG 3. Bode plot of the circuit, showing f_{c1} and f_{c2}

f_{c1}	f_{c2}	f_c	BW
287.437 Hz	17.609 kHz	2.250 kHz	17.322 kHz

Table 6. Data from Multisim

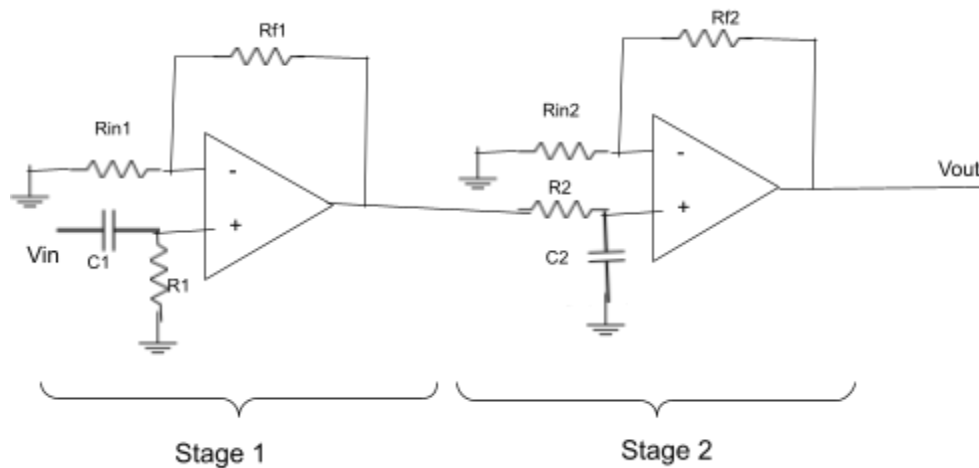
Physical circuit



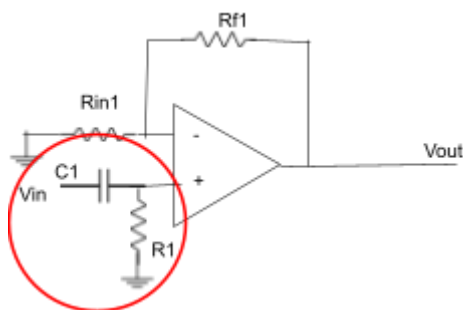
IMG 4. Physical circuit built on breadboard. Note that two operational amplifiers were used but that only one chip is technically needed (see image 1).

Calculations:

First, the transfer function is determined.



For stage 1:



, where $C1=C2=C$ and $R1=2R$

First, we find the transfer function of the passive filter

$$V_{out} = V_{in} \frac{2R}{2R + \frac{1}{CS}}, \quad \frac{V_{out}}{V_{in}} = \frac{2R}{2R + \frac{1}{CS}}$$

We isolate S for stability analysis:

$$\frac{V_{out}}{V_{in}} = \frac{2R}{2R + \frac{1}{CS}} \times \frac{CS}{CS} = \frac{2RCS}{2RCS + 1} = \frac{2RCS}{2RCS + 1} \times \frac{\frac{1}{2RC}}{\frac{1}{2RC}} = \frac{S}{S + \frac{1}{2RC}}$$

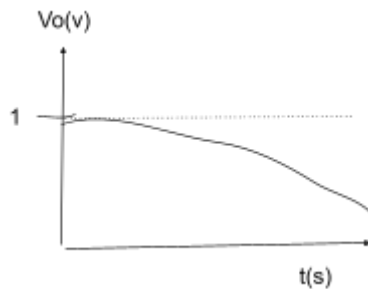
$$H(S) = \frac{V_{out}}{V_{in}}, \text{ and so } H(S) = \frac{S}{S + \frac{1}{2RC}}$$

To check that this is indeed the transfer function of a high pass filter,

$$\text{As } S \rightarrow 0, \quad \frac{S}{S + \frac{1}{2RC}} \approx \frac{(0)}{(0) + \frac{1}{2RC}} \approx 0$$

$$\text{As } S \rightarrow \infty, \quad \frac{S}{S + \frac{1}{2RC}} \approx \frac{(\infty)}{(\infty) + \frac{1}{2RC}} \approx \frac{\infty}{\infty}$$

$$\text{Applying L'Hopital for } \frac{\infty}{\infty} \text{ gives } \left(\frac{\frac{d}{ds}(S)}{\frac{d}{ds}(S + \frac{1}{2RC})} \right) = \left(\frac{1}{1+0} \right), \text{ so } \lim_{s \rightarrow \infty} \left(\frac{1}{1} \right) = 1$$

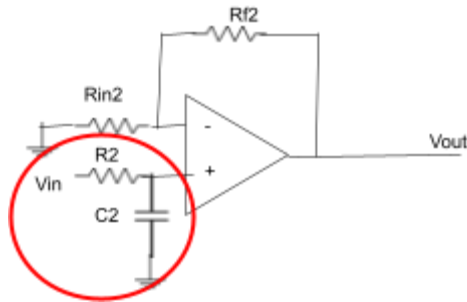


Based on this analysis of the limits of s, the circuit passes low frequency and attenuates high frequency, thus **behaving like a high pass filter**.

The transfer function of stage one is:

$$H(s)_{active (stage 1)} = A_1 \times H(s)_{passive} = 2 \times \frac{S}{S + \frac{1}{2RC}}$$

For stage 2:



, where $R2=R$ and $C1=C2=C$

First, we find the transfer function of the passive filter

$$V_{out} = V_{in} \frac{\frac{1}{CS}}{\frac{1}{CS} + R}, \text{ so } \frac{V_{out}}{V_{in}} = \frac{\frac{1}{CS}}{\frac{1}{CS} + R}.$$

We isolate S for stability analysis:

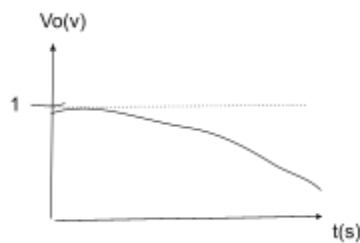
$$\frac{V_{out}}{V_{in}} = \frac{\frac{1}{CS}}{\frac{1}{CS} + R} \times \frac{\frac{1}{RC}}{\frac{1}{RC}} = \frac{\frac{1}{CS RC}}{\frac{1}{CS RC} + \frac{1}{C}}, \frac{V_{out}}{V_{in}} = \frac{\frac{1}{CS RC}}{\frac{1}{CS RC} + \frac{1}{C}} \times \frac{CS}{CS} = \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$$

$$H(S) = \frac{V_{out}}{V_{in}}, H(S) = \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$$

To check that this is indeed the transfer function of a low pass filter,

$$\text{As } S \rightarrow 0, \frac{\frac{1}{RC}}{S + \frac{1}{RC}} \approx \frac{\frac{1}{RC}}{0 + \frac{1}{RC}} \approx \frac{1}{1} \approx 1$$

$$\text{As } S \rightarrow \infty, \frac{\frac{1}{RC}}{S + \frac{1}{RC}} \approx \frac{\frac{1}{RC}}{\infty + \frac{1}{RC}} \approx \frac{1}{\infty} \approx 0$$



Based on this analysis of the limits of s, the circuit passes low frequency and attenuates high frequency, thus **behaving like a low pass filter**.

The transfer function of stage two is:

$$H(s)_{active (stage 2)} = A_2 \times H(s)_{passive} = 2 \times \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$$

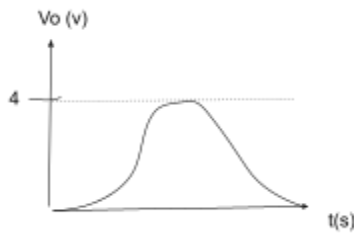
The overall transfer function is

$$\begin{aligned}
H(s)_{total} &= H(s)_{stage 1} \times H(s)_{stage 2} = A_1 \times H(s)_{passive (stage 1)} \times A_2 \times H(s)_{passive (stage 2)} \\
&= 2 \times \frac{S}{S + \frac{1}{2RC}} \times 2 \times \frac{\frac{1}{RC}}{S + \frac{1}{RC}} \\
&= 4 \frac{S \frac{1}{RC}}{(S + \frac{1}{2RC})(S + \frac{1}{RC})} = 4 \frac{S \frac{1}{RC}}{S^2 + \frac{S}{RC} + \frac{S}{2RC} + \frac{1}{2R^2C^2}} = 4 \frac{S \frac{1}{RC}}{S^2 + \frac{3S}{2RC} + \frac{1}{2R^2C^2}} = \frac{4 (S \frac{1}{RC})}{S^2 + (\frac{3}{2RC}) S + \frac{1}{2R^2C^2}} \\
H(s)_{total} &= \frac{4 (S \frac{1}{RC})}{S^2 + (\frac{3}{2RC}) S + \frac{1}{2R^2C^2}}
\end{aligned}$$

To check that this is indeed the transfer function of a band pass filter,

$$\text{As } S \rightarrow 0, H(s)_{total} = \frac{4 (S \frac{1}{RC})}{S^2 + (\frac{3}{2RC}) S + \frac{1}{2R^2C^2}} \approx \frac{4 (0 * \frac{1}{RC})}{0^2 + (\frac{3}{2RC}) (0) + \frac{1}{2R^2C^2}} \approx \frac{0}{\frac{1}{2R^2C^2}} \approx 0$$

$$\text{As } S \rightarrow \infty, H(s)_{total} = \frac{4 (S \frac{1}{RC})}{S^2 + (\frac{3}{2RC}) S + \frac{1}{2R^2C^2}} \approx \frac{4 (\infty * \frac{1}{RC})}{\infty^2 + (\frac{3}{2RC}) (\infty) + \frac{1}{2R^2C^2}} \approx \frac{\infty}{\infty^2} \approx 0$$



Based on this analysis of the limits of s , the circuit attenuates low frequency and attenuates high frequency, thus **behaving like a band pass filter**.

$$H(s)_{total} = \frac{4 (S \frac{1}{RC})}{S^2 + (\frac{3}{2RC}) S + \frac{1}{2R^2C^2}}, \text{ so } N(Z) = 4 (S \frac{1}{RC}) \text{ and } N(P) = S^2 + (\frac{3}{2RC}) S + \frac{1}{2R^2C^2}$$

If $N(Z) = 0$, $0 = 4 (S \frac{1}{RC})$, so $S=0$ or $\frac{1}{RC} = 0$. Therefore there is one zero at zero ($S=0$) and one zero at infinity ($\frac{1}{RC} = 0$)

$$\text{If } N(P) = 0, 0 = S^2 + (\frac{3}{2RC}) S + \frac{1}{2R^2C^2}$$

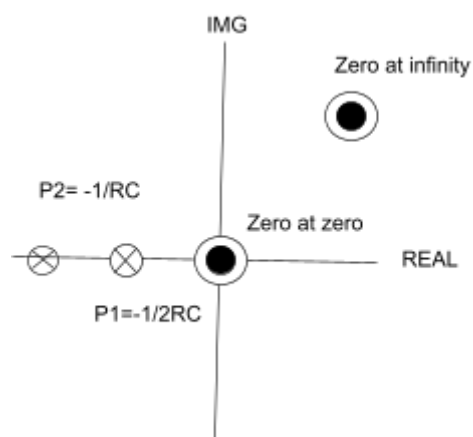
Using the quadratic formula ($\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, where $a = 1$, $b = \frac{3}{2RC}$, $c = \frac{1}{2R^2C^2}$), we get

$$\frac{-\left(\frac{3}{2RC}\right) \pm \sqrt{\left(\frac{3}{2RC}\right)^2 - 4(1)\left(\frac{1}{2R^2C^2}\right)}}{2(1)} = \frac{-\left(\frac{3}{2RC}\right) \pm \sqrt{\frac{9}{4R^2C^2} - \frac{2}{R^2C^2}}}{2} = \frac{-\left(\frac{3}{2RC}\right) \pm \sqrt{\frac{9}{4R^2C^2} - \frac{8}{4R^2C^2}}}{2} = \frac{-\left(\frac{3}{2RC}\right) \pm \sqrt{\frac{1}{4R^2C^2}}}{2}$$

$$= \frac{-\left(\frac{3}{2RC}\right) \pm \frac{1}{\sqrt{4R^2C^2}}}{2} = \frac{-\left(\frac{3}{2RC}\right) \pm \frac{1}{2RC}}{2}$$

$$\text{Pole 1: } S_1 = \frac{-\left(\frac{3}{2RC}\right) + \frac{1}{2RC}}{2} = \frac{-1}{2RC}$$

$$\text{Pole 2: } S_2 = \frac{-\left(\frac{3}{2RC}\right) - \frac{1}{2RC}}{2} = \frac{-1}{RC}$$



To determine if the system is underdamped, critically damped, or over damped we can calculate the quality factor (Q)

$$Q = \frac{fc}{BW} = \frac{fc}{fc_2 - fc_1} = \frac{\sqrt{f_2 f_1}}{f_2 - f_1} = \frac{\sqrt{(1.59k)(3.18k)}}{(3.18k - 1.59k)} = \frac{2248.6}{1590} = 1.41$$

$Q > \frac{1}{2}$	Stable and underdamped
$Q = \frac{1}{2}$	Stable and critically damped
$Q < \frac{1}{2}$	Stable and overdamped

Therefore, the system is stable and underdamped. To check, we can find the dampening ratio

$$\xi = \frac{1}{2Q} = \frac{1}{2.83} = 0.35$$

Underdamped	Critically damped	Overdamped
$0 < \xi < 1$	$\xi = 1$	$\xi > 1$

The dampening ratio confirms that the system is underdamped.

$$\omega_1 = \frac{1}{R1C1} = \frac{1}{2RC} = \frac{1}{2(500\Omega)(0.1\mu F)} = 10,000 \text{ rad/sec}$$

$$\omega_2 = \frac{1}{R2C2} = \frac{1}{RC} = \frac{1}{(500\Omega)(0.1\mu F)} = 20,000 \text{ rad/sec}$$

$$\omega_n = \sqrt{\omega_1 \omega_2} = \sqrt{10,000 \text{ rad/sec} \times 20,000 \text{ rad/sec}} = 14142.14 \text{ rad/sec}$$

	Formula	Calculation	Result
Dampening factor	$\beta = \sqrt{1 - \xi^2}$	$\beta = \sqrt{1 - (0.35)^2}$	$\beta = 0.9367$
	$\omega_d = \beta \omega_n$	$\omega_d = (0.94)(14142.14)$	$\omega_d = 13247 \text{ rad/s}$
Rise time	$T_R = \frac{1+0.7\xi}{\omega_n}$	$T_R = \frac{1+0.7(0.35)}{14142.14}$	$T_R = 0.088 \text{ ms}$
Delay time	$T_D = \frac{0.8+2.5\xi}{\omega_n}$	$T_D = \frac{0.8+2.5(0.35)}{14142.14}$	$T_D = 0.1184 \text{ ms}$
Peak time	$T_p = \frac{\pi}{\omega_d}$	$T_p = \frac{\pi}{0.2374}$	$T_p = 0.237 \text{ ms}$
% Overshoot	$O.S\% = e^{\frac{-\pi\xi}{\beta}} * 100$	$O.S\% = e^{\frac{-\pi(0.35)}{0.94}} * 100$	$O.S\% = 30.9\%$
Settling time	$t_s = \frac{-\ln(\% \text{ in decimal})}{\xi \omega_n}$	For 2%: $t_s = \frac{-\ln(0.02)}{(0.35)(10,000)}$	$t_s = 0.790 \text{ ms}$
		For 4%: $t_s = \frac{-\ln(0.04)}{(0.35)(10,000)}$	$t_s = 0.650 \text{ ms}$
		For 5%: $t_s = \frac{-\ln(0.05)}{(0.35)(10,000)}$	$t_s = 0.605 \text{ ms}$

Table 7. Underdamped system calculated values

The step response for an underdamped system is $V_o(t) = 1 - e^{-\beta t}(\cos(\omega_d t) + \frac{\xi}{\beta} \sin(\omega_d t))$

$$V_o(t) = 1 - e^{-(0.94)t}(\cos 13246.8t) + \frac{0.35}{0.94} \sin(13246.8t)$$

From the standard underdamped form, $e^{-\beta t} \sin \omega_d t = e^{-\frac{t}{\tau}} \sin \omega_d t$ and $\tau = \frac{1}{\beta} = \frac{1}{0.94} = 1.068 \text{ s}$

For 5τ , $5\tau = 5(1.068 \text{ s}) = 5.3380 \text{ s}$

$$\text{At } t = 5\tau, e^{-\beta t} = e^{-\beta(5\tau)} = e^{-\beta(5 \frac{1}{\beta})} = e^{-5}$$

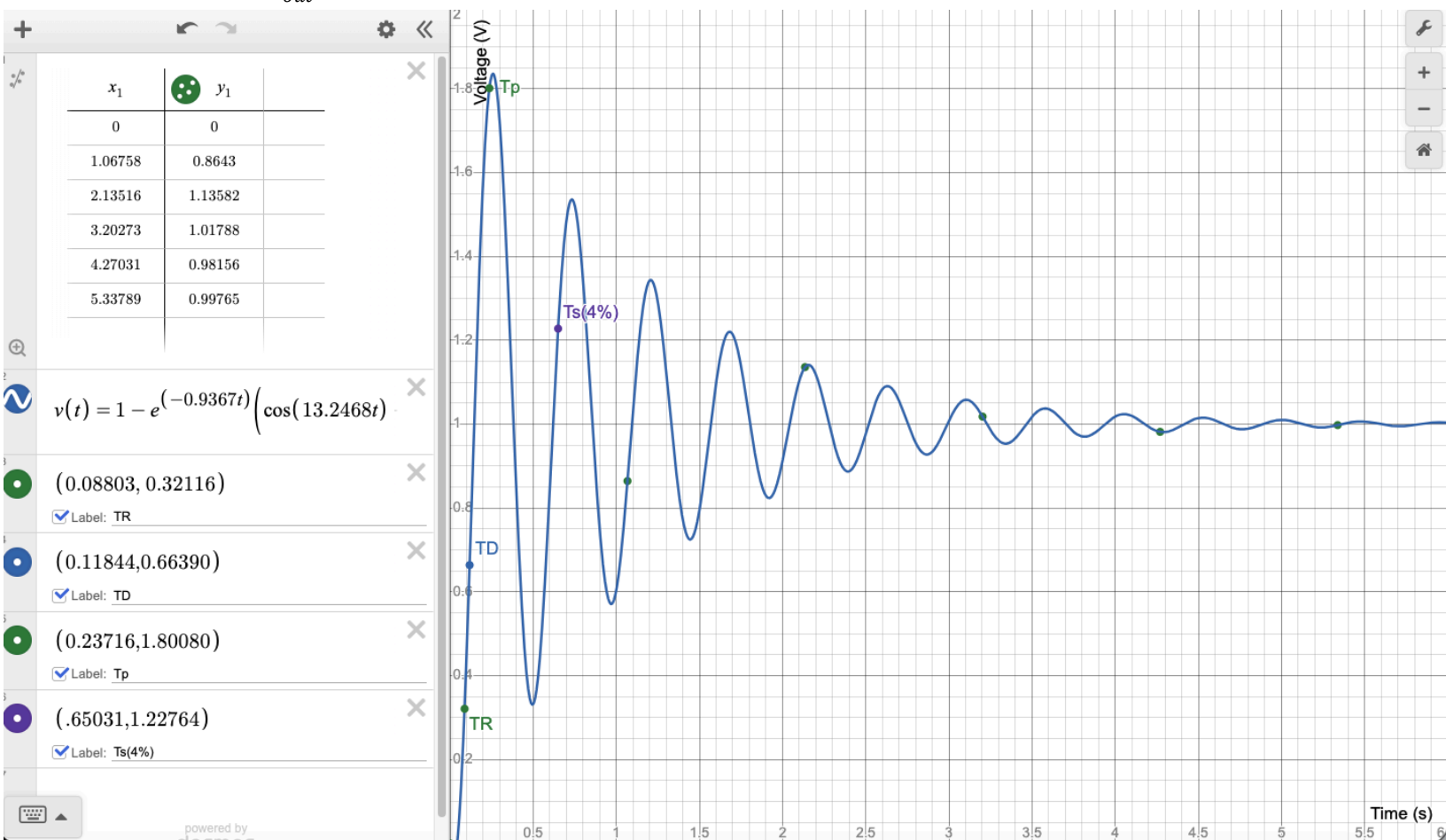
$$V_o(t) = 1 - e^{-5}(\cos 13246.8t) + \frac{0.35}{0.94} \sin(13246.8t)$$

$$V_o(t) = 1 - 0.0067370(\cos(13246.8 * 1.068) + 0.3737\sin(13246.8*1.068))=0.9987v$$

To find the % of the final value that the output is, $\frac{0.9987v}{1v} * 100 = 99.87\%$

	$V_{out}(t)$	% Final
$0\tau=0s$	$0v$	0%
$1\tau = 1.0676s$	$0.8643v$	109.68%
$2\tau = 2.1351s$	$1.1358v$	114.30%
$3\tau = 3.2027s$	$1.0179v$	99.83%
$4\tau = 4.2703s$	$0.9816v$	98.05%
$5\tau = 5.3379s$	$0.9977v$	99.87%

Table 8. $V_{out}(t)$ after time constants



IMG 6.

Steady state is reached after 5τ , graphed using [Desmos.com](https://www.desmos.com). Note that the equation of the graph was divided by 1000 to decrease the frequency to make the points visible with the line of best fit. This works because reading in milliseconds stretches the graph horizontally. The equation used in this graph is $V_o(t) = 1 - e^{-(0.94)t}(\cos 13.2468t) + \frac{0.35}{0.94}\sin(13.2468t)$ instead of $V_o(t) = 1 - e^{-(0.94)t}(\cos 13246.8t) + \frac{0.35}{0.94}\sin(13246.8t)$

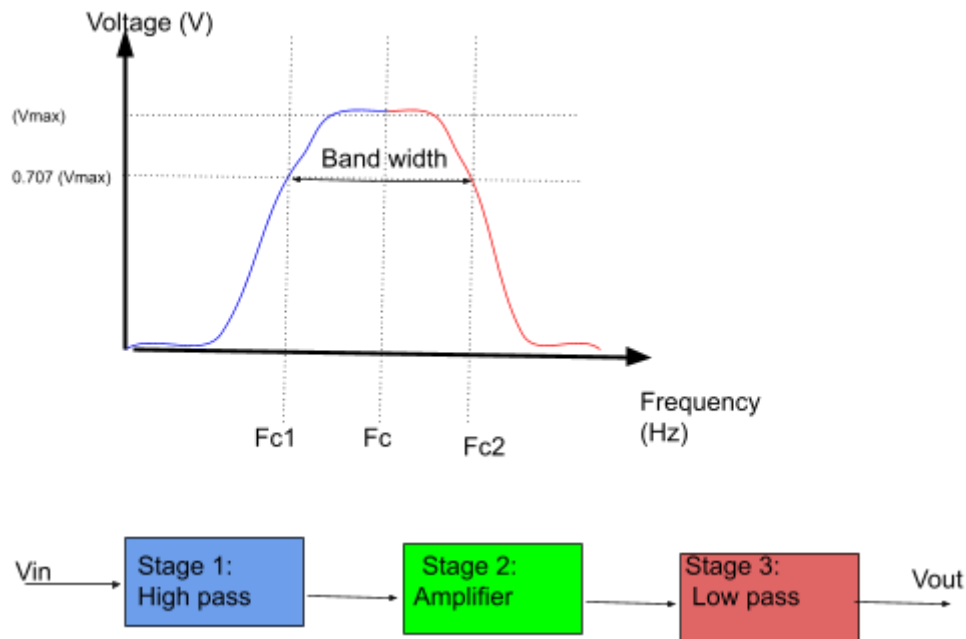
Results:

	fc1	fc2	fc	BW
Calculated	1.59 kHz	3.18 kHz	2.249 kHz	1.590 kHz
Measured	287.437 Hz	17.609 kHz	2.250 kHz	17.322 kHz

Table 9. *Calculated verses measured (Multisim)*

For configuration 1, as seen in table 9, the calculated center frequency and measured almost match. This confirms that the component values were appropriately selected and the circuit is functioning as intended. However, the variations of the passband (fc1, fc2 and the bandwidth) have a large discrepancy. Our hypothesis is that this is due to the peak voltage being amplified by the total gain of both stages combined (4). Because the Bode plot magnitude is elevated due to the gain, the -3dB point appear shifted thus widening the bandwidth.

BAND PASS FILTER, SECOND CONFIGURATION: *A second order non inverting band pass filter with three stages- a high pass filter, an amplifier and a low pass filter. The operational amplifier has a gain and does not act as a buffer.*



1.Determining component values

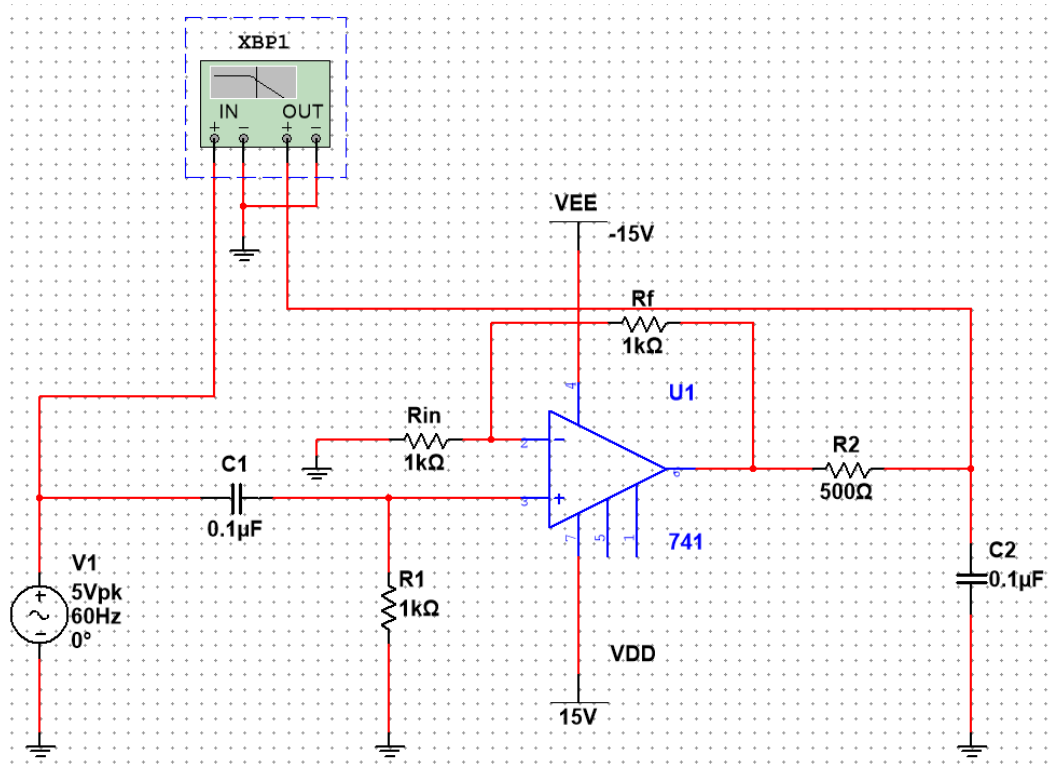
Since $F_{c1} = 1.59 \text{ kHz}$ and $F_{c2} = 3.18 \text{ kHz}$, we can keep the same component values as the previous configuration. However, in this configuration there is only one op amp and therefore one gain, and one R_f and R_{in} .

For a gain of 2, $A_v = 1 + \frac{R_f}{R_{in}} = 1 + \frac{1 \text{ k}\Omega}{1 \text{ k}\Omega} = 2$

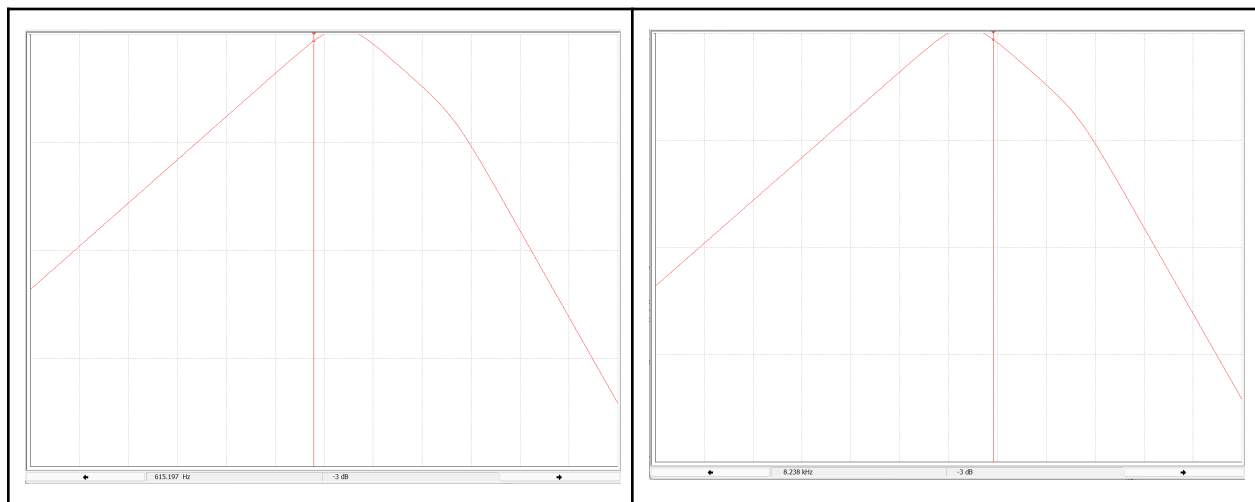
	Circuit from Multisim	Circuit built on breadboard
C=C1=C2	0.1uF	0.1uF
R1	R1=2R=1000	820+180=1k
R2	R2=R=500	470+33=503
R_{in}	1k Ω	1k Ω
R_f	1k Ω	1k Ω
Gain	2	2

Table 10. *Component values*

Multisim



IMG 7. Schematic of circuit on Multisim



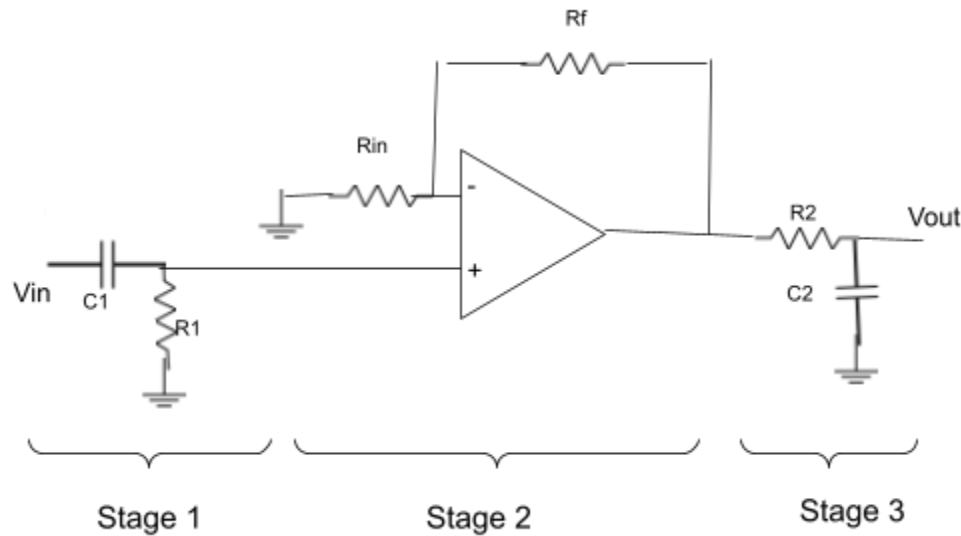
IMG 8. Bode plot of the circuit, showing $fc1$ and $fc2$

fc1	fc2	fc	BW
615.197 Hz	8.238 kHz	2.251 kHz	7.623 kHz

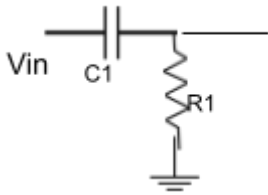
Table 11. Data from Multisim

Calculations:

First, the transfer function is determined.



For stage 1:



, where $C1=C2=C$ and $R1=2R$

First, we find the transfer function of the passive filter

$$V_{out} = V_{in} \frac{2R}{2R + \frac{1}{CS}}, \quad \frac{V_{out}}{V_{in}} = \frac{2R}{2R + \frac{1}{CS}}$$

We isolate S for stability analysis:

$$\frac{V_{out}}{V_{in}} = \frac{2R}{2R + \frac{1}{CS}} \times \frac{CS}{CS} = \frac{2RCS}{2RCS + 1} = \frac{2RCS}{2RCS + 1} \times \frac{\frac{1}{2RC}}{\frac{1}{2RC}} = \frac{S}{S + \frac{1}{2RC}}$$

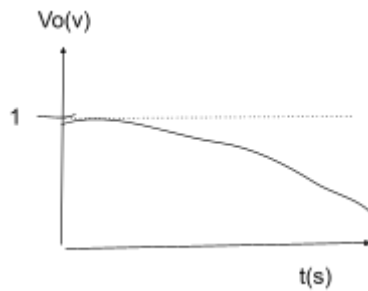
$$H(S) = \frac{V_{out}}{V_{in}}, \text{ and so } H(S) = \frac{S}{S + \frac{1}{2RC}}$$

To check that this is indeed the transfer function of a high pass filter,

$$\text{As } S \rightarrow 0, \quad \frac{S}{S + \frac{1}{2RC}} \approx \frac{(0)}{(0) + \frac{1}{2RC}} \approx 0$$

$$\text{As } S \rightarrow \infty, \quad \frac{S}{S + \frac{1}{2RC}} \approx \frac{(\infty)}{(\infty) + \frac{1}{2RC}} \approx \frac{\infty}{\infty}$$

Applying L'Hopital for $\frac{\infty}{\infty}$ gives $\left(\frac{\frac{d}{ds}(S)}{\frac{d}{ds}(S+\frac{1}{2RC})}\right)=\left(\frac{1}{1+0}\right)$, so $\lim_{s \rightarrow \infty} \left(\frac{1}{1}\right)=1$

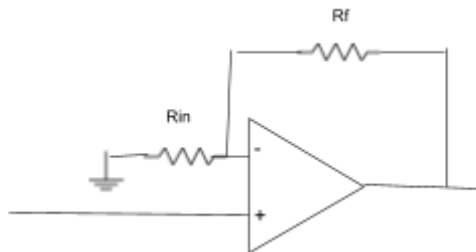


Based on this analysis of the limits of s , the circuit passes low frequency and attenuates high frequency, thus **behaving like a high pass filter**.

The transfer function of stage one is:

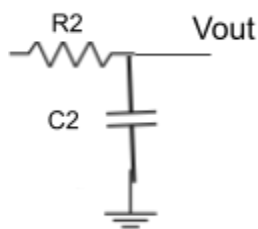
$$H(s)_{passive (stage 1)} = \frac{S}{S + \frac{1}{2RC}}$$

For stage 2:



It was previously calculated that we have a gain of 2, since $Av = 1 + \frac{Rf}{Rin} = 1 + \frac{1k\Omega}{1k\Omega} = 2$

For stage 3:



, where $R2=R$ and $C1=C2=C$

First, we find the transfer function of the passive filter

$$V_{out} = V_{in} \frac{\frac{1}{CS}}{\frac{1}{CS} + R}, \text{ so } \frac{V_{out}}{V_{in}} = \frac{\frac{1}{CS}}{\frac{1}{CS} + R}.$$

We isolate S for stability analysis:

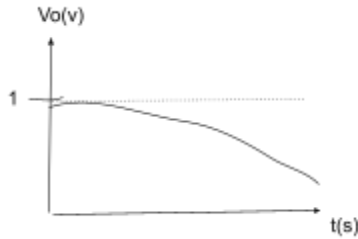
$$\frac{V_{out}}{V_{in}} = \frac{\frac{1}{CS}}{\frac{1}{CS} + R} \times \frac{\frac{1}{RC}}{\frac{1}{RC}} = \frac{\frac{1}{CS RC}}{\frac{1}{CS RC} + \frac{1}{C}}, \frac{V_{out}}{V_{in}} = \frac{\frac{1}{CS RC}}{\frac{1}{CS RC} + \frac{1}{C}} \times \frac{CS}{CS} = \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$$

$$H(S) = \frac{V_{out}}{V_{in}}, H(S) = \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$$

To check that this is indeed the transfer function of a low pass filter,

$$\text{As } S \rightarrow 0, \frac{\frac{1}{RC}}{S + \frac{1}{RC}} \approx \frac{\frac{1}{RC}}{0 + \frac{1}{RC}} \approx \frac{1}{1} \approx 1$$

$$\text{As } S \rightarrow \infty, \frac{\frac{1}{RC}}{S + \frac{1}{RC}} \approx \frac{\frac{1}{RC}}{\infty + \frac{1}{RC}} \approx \frac{1}{\infty} \approx 0$$



Based on this analysis of the limits of s, the circuit passes low frequency and attenuates high frequency, thus **behaving like a low pass filter**.

The transfer function of stage three is:

$$H(s)_{passive (stage 3)} = \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$$

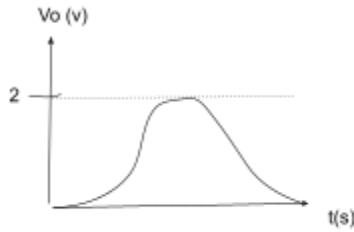
The overall transfer function is

$$\begin{aligned} H(s)_{total} &= H(s)_{stage 1} \times H(s)_{stage 2} \times H(s)_{stage 3} = \frac{S}{S + \frac{1}{2RC}} \times 2 \times \frac{\frac{1}{RC}}{S + \frac{1}{RC}} \\ &= 2 \frac{S \frac{1}{RC}}{(S + \frac{1}{2RC})(S + \frac{1}{RC})} = 2 \frac{S \frac{1}{RC}}{S^2 + \frac{S}{RC} + \frac{S}{2RC} + \frac{1}{2R^2C^2}} = 2 \frac{S \frac{1}{RC}}{S^2 + \frac{3S}{2RC} + \frac{1}{2R^2C^2}} = \frac{2(S \frac{1}{RC})}{S^2 + (\frac{3}{2RC})S + \frac{1}{2R^2C^2}} \\ H(s)_{total} &= \frac{2(S \frac{1}{RC})}{S^2 + (\frac{3}{2RC})S + \frac{1}{2R^2C^2}} \end{aligned}$$

To check that this is indeed the transfer function of a band pass filter,

$$\text{As } S \rightarrow 0, H(s)_{total} = \frac{2(S \frac{1}{RC})}{S^2 + (\frac{3}{2RC})S + \frac{1}{2R^2C^2}} \approx \frac{2(0 * \frac{1}{RC})}{0^2 + (\frac{3}{2RC})(0) + \frac{1}{2R^2C^2}} \approx \frac{0}{\frac{1}{2R^2C^2}} \approx 0$$

$$\text{As } S \rightarrow \infty, H(s)_{total} = \frac{2(S \frac{1}{RC})}{S^2 + (\frac{3}{2RC})S + \frac{1}{2R^2C^2}} \approx \frac{2(\infty * \frac{1}{RC})}{\infty^2 + (\frac{3}{2RC})(\infty) + \frac{1}{2R^2C^2}} \approx \frac{\infty}{\infty^2} \approx 0$$



Based on this analysis of the limits of s, the circuit attenuates low frequency and attenuates high frequency, thus **behaving like a band pass filter**.

$$H(s)_{total} = \frac{2(S \frac{1}{RC})}{S^2 + (\frac{3}{2RC})S + \frac{1}{2R^2C^2}}, \text{ so } N(Z) = 2(S \frac{1}{RC}) \text{ and } N(P) = S^2 + (\frac{3}{2RC})S + \frac{1}{2R^2C^2}$$

If $N(Z) = 0$ then $0 = 2(S \frac{1}{RC})$, so $S=0$ or $\frac{1}{RC} = 0$. Therefore there is one zero at zero ($S=0$) and one zero at infinity ($\frac{1}{RC} = 0$), indicating band pass filter behavior.

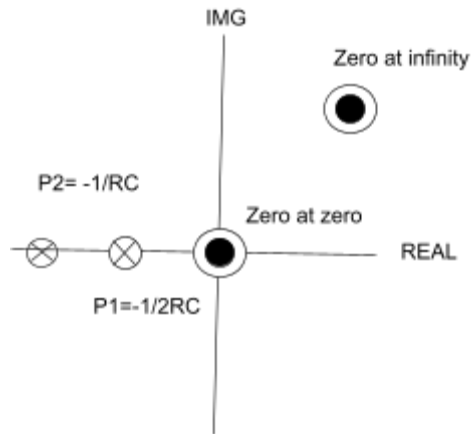
$$\text{If } N(P) = 0, 0 = S^2 + (\frac{3}{2RC})S + \frac{1}{2R^2C^2}$$

Using the quadratic formula ($\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, where $a = 1, b = \frac{3}{2RC}, c = \frac{1}{2R^2C^2}$), we get

$$\begin{aligned} \frac{-\left(\frac{3}{2RC}\right) \pm \sqrt{\left(\frac{3}{2RC}\right)^2 - 4(1)\left(\frac{1}{2R^2C^2}\right)}}{2(1)} &= \frac{-\left(\frac{3}{2RC}\right) \pm \sqrt{\frac{9}{4R^2C^2} - \frac{2}{R^2C^2}}}{2} = \frac{-\left(\frac{3}{2RC}\right) \pm \sqrt{\frac{9}{4R^2C^2} - \frac{8}{4R^2C^2}}}{2} = \frac{-\left(\frac{3}{2RC}\right) \pm \sqrt{\frac{1}{4R^2C^2}}}{2} \\ &= \frac{-\left(\frac{3}{2RC}\right) \pm \frac{1}{\sqrt{4R^2C^2}}}{2} = \frac{-\left(\frac{3}{2RC}\right) \pm \frac{1}{2RC}}{2} \end{aligned}$$

$$\text{Pole 1: } S_1 = \frac{-\left(\frac{3}{2RC}\right) + \frac{1}{2RC}}{2} = \frac{-1}{2RC}$$

$$\text{Pole 2: } S_2 = \frac{-\left(\frac{3}{2RC}\right) - \frac{1}{2RC}}{2} = \frac{-1}{RC}$$



As just the gain in the transfer function differs from the circuit from the first configuration, we have the same poles and zeros.

To determine if the system is underdamped, critically damped, or over damped we can calculate the quality factor (Q)

$$Q = \frac{fc}{BW} = \frac{fc}{fc_2 - fc_1} = \frac{\sqrt{f_2 f_1}}{f_2 - f_1} = \frac{\sqrt{(1.59k)(3.18k)}}{(3.18k - 1.59k)} = \frac{2248.6}{1590} = 1.41$$

$Q > \frac{1}{2}$	Stable and underdamped
$Q = \frac{1}{2}$	Stable and critically dampened
$Q < \frac{1}{2}$	Stable and overdamped

Therefore, the system is stable and underdamped. To check, we can find the dampening ratio

$$\xi = \frac{1}{2Q} = \frac{1}{2.83} = 0.35$$

Underdamped	Critically damped	Overdamped
$0 < \xi < 1$	$\xi = 1$	$\xi > 1$

The dampening ratio confirms that the system is underdamped. As just the gain changed from the circuit of configuration 1 compared to this circuit, we also have the same underdamped characteristics and same step response. Therefore, the mathematical work is identical, and so the results sans process is shown below. **For the calculation process, please refer to the calculation process of the first circuit configuration on page 19.**

$$\omega_1 = 10,000 \text{ rad/sec}$$

$$\omega_2 = 20,000 \text{ rad/sec}$$

$$\omega_n = 14142.14 \text{ rad/sec}$$

	Formula	Result
Dampening factor	$\beta = \sqrt{1 - \xi^2}$	$\beta = 0.9367$
	$\omega_d = \beta\omega_n$	$\omega_d = 13247 \text{ rad/s}$
Rise time	$T_R = \frac{1+0.7\xi}{\omega_n}$	$T_R = 0.088\text{ms}$
Delay time	$T_D = \frac{0.8+2.5\xi}{\omega_n}$	$T_D = 0.1184\text{ms}$
Peak time	$T_p = \frac{\pi}{\omega_d}$	$T_p = 0.237\text{ms}$
% Overshoot	$O.S\% = e^{\frac{-\pi\xi}{\beta}} * 100$	$O.S\% = 30.9\%$
Settling time	$t_s = \frac{-\ln(\% \text{ in decimal})}{\xi\omega_n}$	2%: $t_s = 0.790 \text{ ms}$
		4%: $t_s = 0.650 \text{ ms}$
		5%: $t_s = 0.605 \text{ ms}$

Table 12.

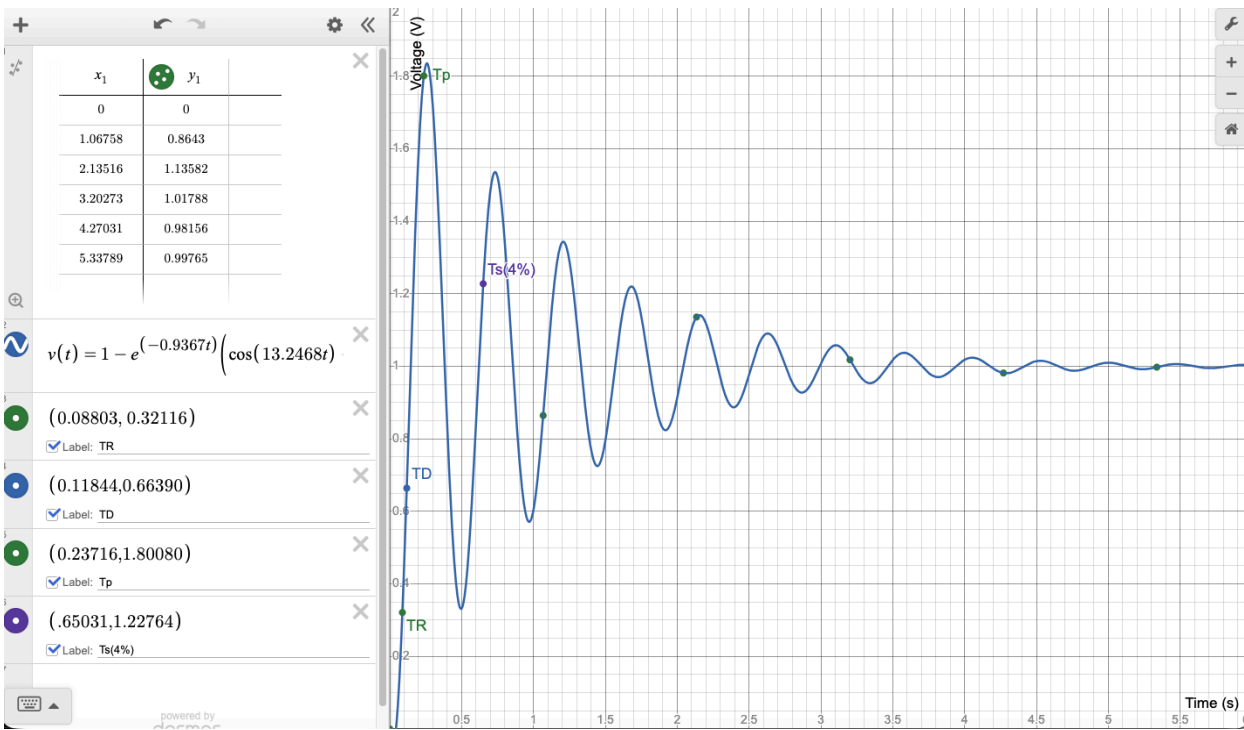
The step response for this underdamped system is still

$$V_o(t) = 1 - e^{-(0.94)t} (\cos 13246.8t) + \frac{0.35}{0.94} \sin(13246.8t).$$

	$V_{out}(t)$	% Final
$0\tau=0\text{s}$	0v	0%
$1\tau = 1.0676\text{s}$	0.8643vv	109.68%
$2\tau = 2.1351\text{s}$	1.1358v	114.30%
$3\tau = 3.2027\text{s}$	1.0179v	99.83%

$4\tau = 4.2703s$	0.9816v	98.05%
$5\tau = 5.3379s$	0.9977v	99.87%

Table 13. $V_{out}(t)$ after time constants



IMG 9. Please note that the step response is the same as the previous circuit. The same underdamped, oscillatory behavior with the same aspects can be observed.

Steady state is reached after 5τ , graphed using [Desmos.com](https://www.desmos.com). Note that the equation of the graph was divided by 1000 to decrease the frequency to make the points visible with the line of best fit.

This works because reading in milliseconds stretches the graph horizontally. The equation used

in this graph is $V_o(t) = 1 - e^{-(0.94)t}(\cos 13.2468t) + \frac{0.35}{0.94}\sin(13.2468t)$ instead of

$$V_o(t) = 1 - e^{-(0.94)t}(\cos 13246.8t) + \frac{0.35}{0.94}\sin(13246.8t)$$

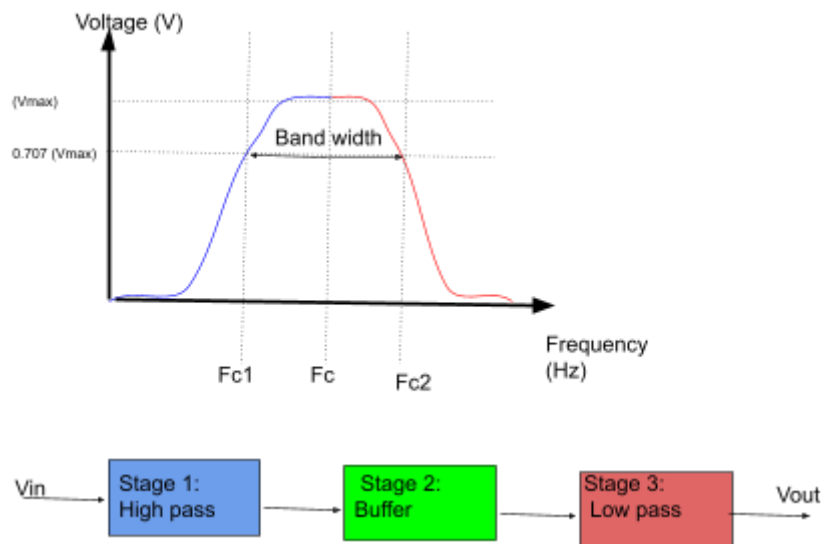
Results:

	fc1	fc2	fc	BW
calculated	1.59 kHz	3.18 kHz	2.249 kHz	1.590 kHz
measured	615.197 Hz	8.238 kHz	2.251 kHz	7.623 kHz

Table 14

As seen in the measured and calculated results for Configuration 2, the center frequency values are very close. This suggests that the selected component values were appropriate and that the circuit functioned generally as a band-pass filter, confirmed by the Bode plot shape and calculations. However, the measured lower and upper cutoff frequencies as well as bandwidth do vary compared to the calculated values. This indicates that the simulated passband was much broader than expected (or calculated). One possible explanation is that the amplifier stage, with gain 2, raised the passband magnitude and influenced where the -3 dB cutoff points were seen on the Bode plot. Although the passband edges varied significantly, the close agreement in center frequency shows that the circuit still operated near the intended center of the design.

BAND PASS FILTER, THIRD CONFIGURATION: *A second-order non-inverting band pass filter with three stages- a high pass filter, an amplifier, and a low pass filter. The operational amplifier does not have a gain and acts as a buffer.*



1.Determining component values

Since $F_{c1} = 1.59kHz$ and $F_{c2} = 3.18kHz$, we can keep the same component values for R1,

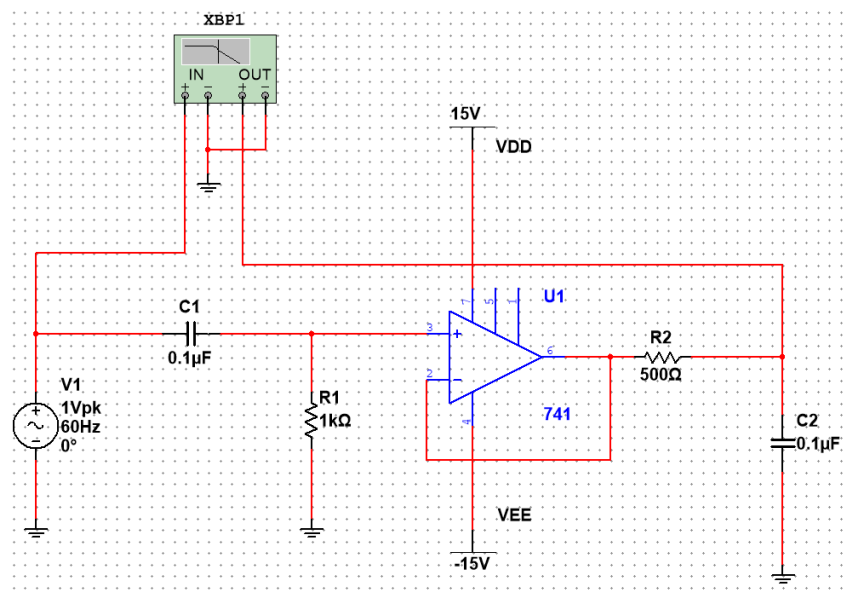
R2, C1 and C2 as the previous configuration. However, in this configuration the op amp acts as a buffer, meaning that Rf is 0 to meet the condition the gain being 1. To make the gain be 1,

$$Av = 1 = 1 + \frac{R_f}{R_{in}}, \text{ so } \frac{R_f}{R_{in}} = 0. \text{ Therefore, } R_{in} \text{ is also equal to } 0.$$

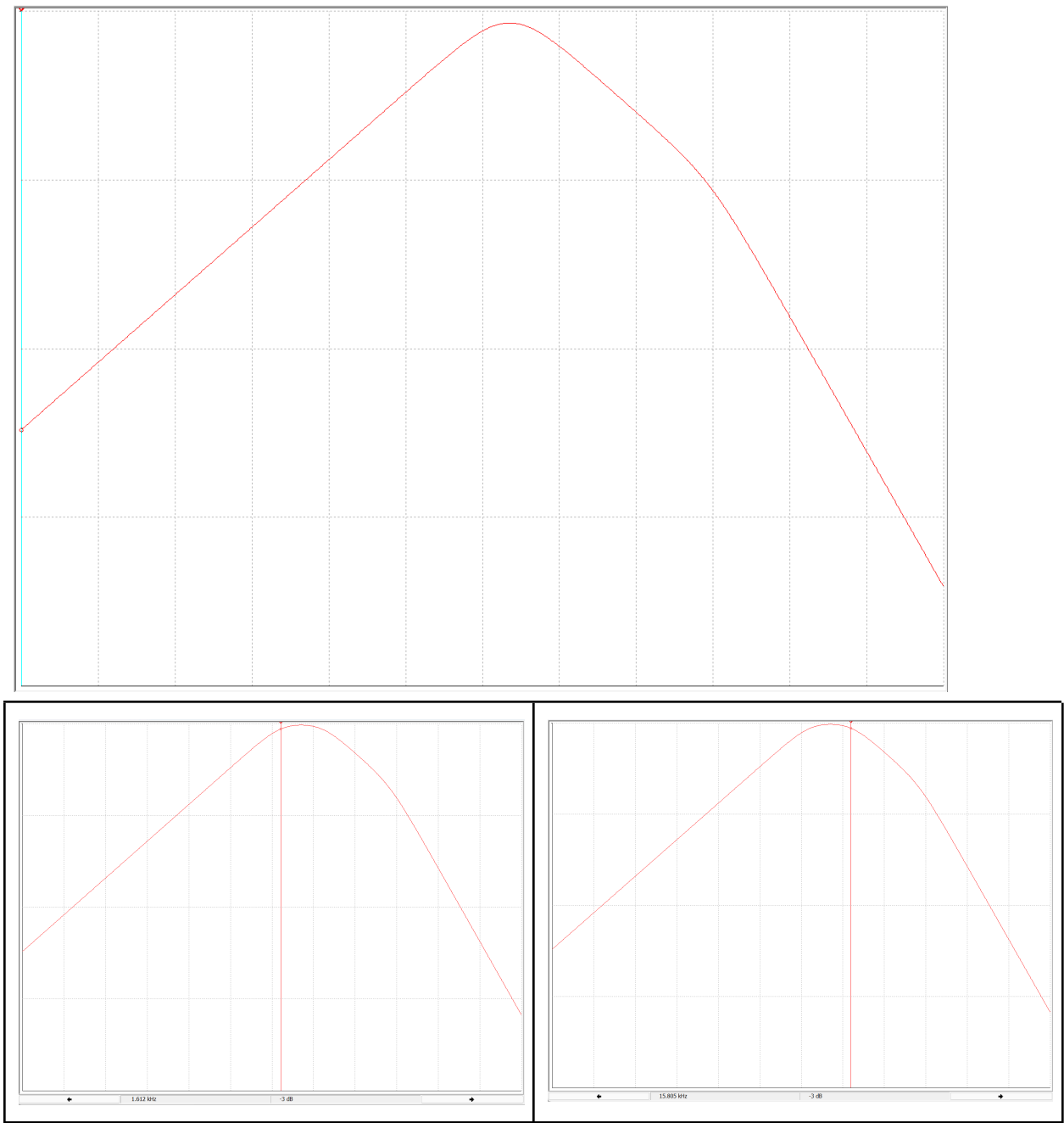
	Circuit from Multisim	Circuit built on breadboard
C=C1=C2	0.1uF	0.1uF
R1	R1=2R=1000	820+180=1k
R2	R2=R=500	470+33=503
Gain	1	1

Table 15. Component values

Multisim



IMG 10. Schematic of circuit from Multisim

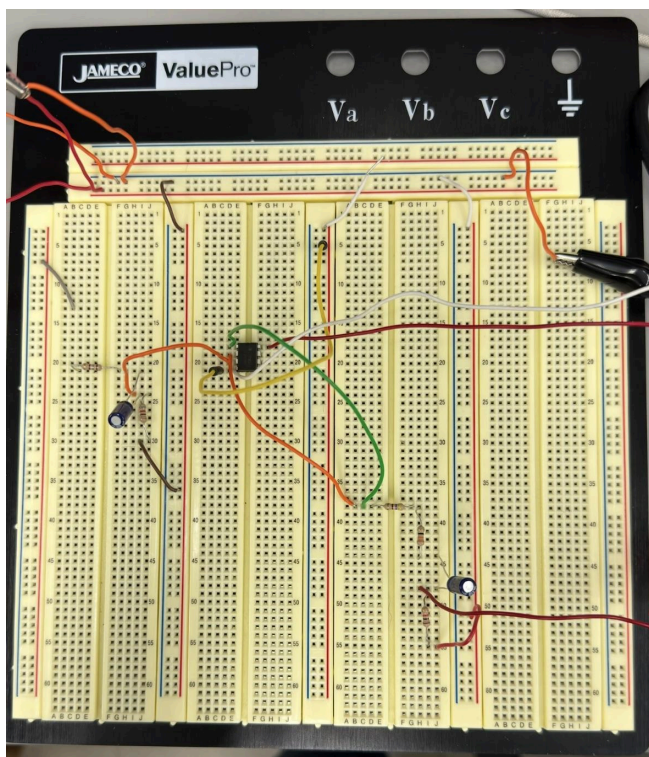


IMG 11. Bode plot of the circuit

fc1	fc2	fc	BW
1.612 kHz	15.805 kHz	5.048 kHz	14.193 kHz

Table 16. Data from Multisim

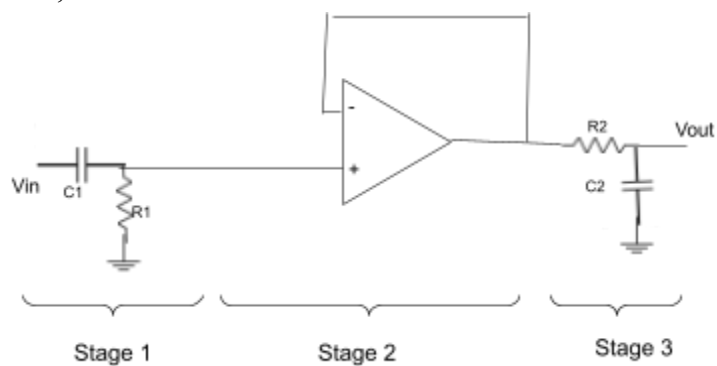
Physical circuit



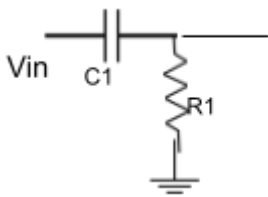
IMG 12. *physical circuit*

Calculations:

First, the transfer function is determined.



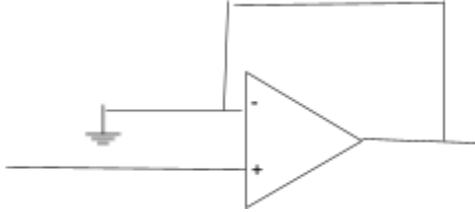
For stage 1:



, where $C1=C2=C$ and $R1=2R$

First, we find the transfer function of the passive filter. This was previously found and proven to be $H(s)_{passive (stage 1)} = \frac{S}{S + \frac{1}{2RC}}$. **For the process and calculations, please see page 23.**

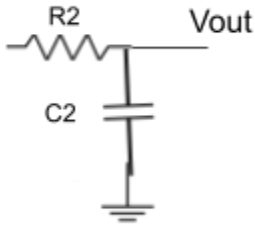
For stage 2:



It was previously determined that we have a gain of 1 for a unity gain buffer, since

$$Av = 1 + \frac{R_f}{R_{in}} = 1 + 0 = 1$$

For stage 3:



, where $R2=R$ and $C1=C2=C$

The transfer function was previously proven to be $H(s)_{passive (stage 3)} = \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$. **For the process and calculations, please see page 26.**

The overall transfer function is

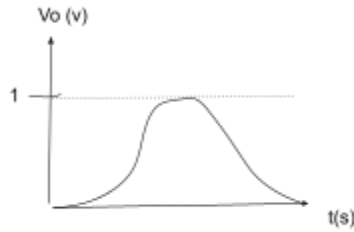
$$H(s)_{total} = H(s)_{stage 1} \times H(s)_{stage 2} \times H(s)_{stage 3} = \frac{S}{S + \frac{1}{2RC}} \times 1 \times \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$$

$$H(s)_{total} = \frac{(S \frac{1}{RC})}{S^2 + (\frac{3}{2RC})S + \frac{1}{2R^2C^2}}$$

To check that this is indeed the transfer function of a band pass filter,

$$\text{As } S \rightarrow 0, H(s)_{total} = \frac{(S \frac{1}{RC})}{S^2 + (\frac{3}{2RC})S + \frac{1}{2R^2C^2}} \approx \frac{(0 * \frac{1}{RC})}{0^2 + (\frac{3}{2RC})(0) + \frac{1}{2R^2C^2}} \approx \frac{0}{\frac{1}{2R^2C^2}} \approx 0$$

$$\text{As } S \rightarrow \infty, H(s)_{total} = \frac{(S \frac{1}{RC})}{S^2 + (\frac{3}{2RC})S + \frac{1}{2R^2C^2}} \approx \frac{(\infty * \frac{1}{RC})}{\infty^2 + (\frac{3}{2RC})(\infty) + \frac{1}{2R^2C^2}} \approx \frac{\infty}{\infty^2} \approx 0$$



Based on this analysis of the limits of s , the circuit attenuates low frequency and attenuates high frequency, thus **behaving like a band pass filter**.

$$H(s)_{total} = \frac{(S \frac{1}{RC})}{S^2 + (\frac{3}{2RC})S + \frac{1}{2R^2C^2}}, \text{ so } N(Z) = (S \frac{1}{RC}) \text{ and } N(P) = S^2 + (\frac{3}{2RC})S + \frac{1}{2R^2C^2}$$

If $N(Z) = 0$ then $0 = (S \frac{1}{RC})$, so $S=0$ or $\frac{1}{RC} = 0$. Therefore there is one zero at zero ($S=0$) and one zero at infinity ($\frac{1}{RC} = 0$), indicating band pass filter behavior. This remains the same compared to the other two configurations previously covered, as only the gain changed.

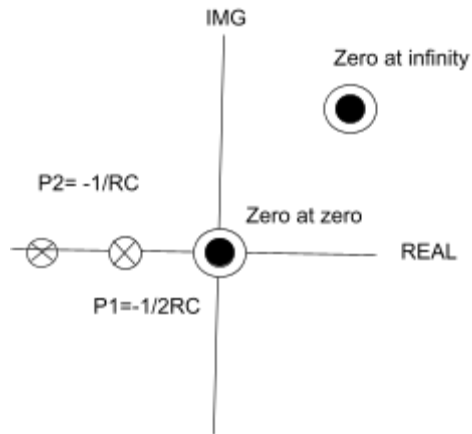
$$\text{If } N(P) = 0, 0 = S^2 + (\frac{3}{2RC})S + \frac{1}{2R^2C^2}$$

Using the quadratic formula ($\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, where $a = 1$, $b = \frac{3}{2RC}$, $c = \frac{1}{2R^2C^2}$), we get

$$\begin{aligned} \frac{-\left(\frac{3}{2RC}\right) \pm \sqrt{\left(\frac{3}{2RC}\right)^2 - 4(1)\left(\frac{1}{2R^2C^2}\right)}}{2(1)} &= \frac{-\left(\frac{3}{2RC}\right) \pm \sqrt{\frac{9}{4R^2C^2} - \frac{2}{R^2C^2}}}{2} = \frac{-\left(\frac{3}{2RC}\right) \pm \sqrt{\frac{9}{4R^2C^2} - \frac{8}{4R^2C^2}}}{2} = \frac{-\left(\frac{3}{2RC}\right) \pm \sqrt{\frac{1}{4R^2C^2}}}{2} \\ &= \frac{-\left(\frac{3}{2RC}\right) \pm \frac{1}{\sqrt{4R^2C^2}}}{2} = \frac{-\left(\frac{3}{2RC}\right) \pm \frac{1}{2RC}}{2} \end{aligned}$$

$$\text{Pole 1: } S_1 = \frac{-\left(\frac{3}{2RC}\right) + \frac{1}{2RC}}{2} = \frac{-1}{2RC}$$

$$\text{Pole 2: } S_2 = \frac{-\left(\frac{3}{2RC}\right) - \frac{1}{2RC}}{2} = \frac{-1}{RC}$$



As just the gain in the transfer function differs from the circuit from the first configuration, we have the same poles and zeros.

To determine if the system is underdamped, critically dampened, or over dampened we can calculate the quality factor (Q)

$$Q = \frac{f_c}{BW} = \frac{f_c}{f_{c_2} - f_{c_1}} = \frac{\sqrt{f_2 f_1}}{f_2 - f_1} = \frac{\sqrt{(1.59k)(3.18k)}}{(3.18k - 1.59k)} = \frac{2248.6}{1590} = 1.41$$

$Q > \frac{1}{2}$	Stable and underdamped
$Q = \frac{1}{2}$	Stable and critically dampened
$Q < \frac{1}{2}$	Stable and overdamped

Therefore, the system is stable and underdamped. To check, we can find the dampening ratio

$$\xi = \frac{1}{2Q} = \frac{1}{2.83} = 0.35$$

Underdamped	Critically dampened	Overdamped
$0 < \xi < 1$	$\xi = 1$	$\xi > 1$

The dampening ratio confirms that the system is underdamped. As just the gain changed, we still have the same underdamped characteristics and the same step response compared to the configurations in the first two circuits. Therefore, the mathematical work is identical, and so the results sans process is shown below. **For the calculation process, please refer to the calculation process of the first circuit configuration on page 19.**

$\omega_1 = 10,000 \text{ rad/sec}$ and $\omega_2 = 20,000 \text{ rad/sec}$, so $\omega_n = 14142.14 \text{ rad/sec}$

	Formula	Result
Dampening factor	$\beta = \sqrt{1 - \xi^2}$	$\beta = 0.9367$
	$\omega_d = \beta\omega_n$	$\omega_d = 13247 \text{ rad/s}$
Rise time	$T_R = \frac{1 + 0.7\xi}{\omega_n}$	$T_R = 0.088 \text{ ms}$
Delay time	$T_D = \frac{0.8 + 2.5\xi}{\omega_n}$	$T_D = 0.1184 \text{ ms}$
Peak time	$T_p = \frac{\pi}{\omega_d}$	$T_p = 0.237 \text{ ms}$
% Overshoot	$O.S\% = e^{\frac{-\pi\xi}{\beta}} * 100$	$O.S\% = 30.9\%$
Settling time	$t_s = \frac{-\ln(\% \text{ in decimal})}{\xi\omega_n}$	2%: $t_s = 0.790 \text{ ms}$
		4%: $t_s = 0.650 \text{ ms}$
		5%: $t_s = 0.605 \text{ ms}$

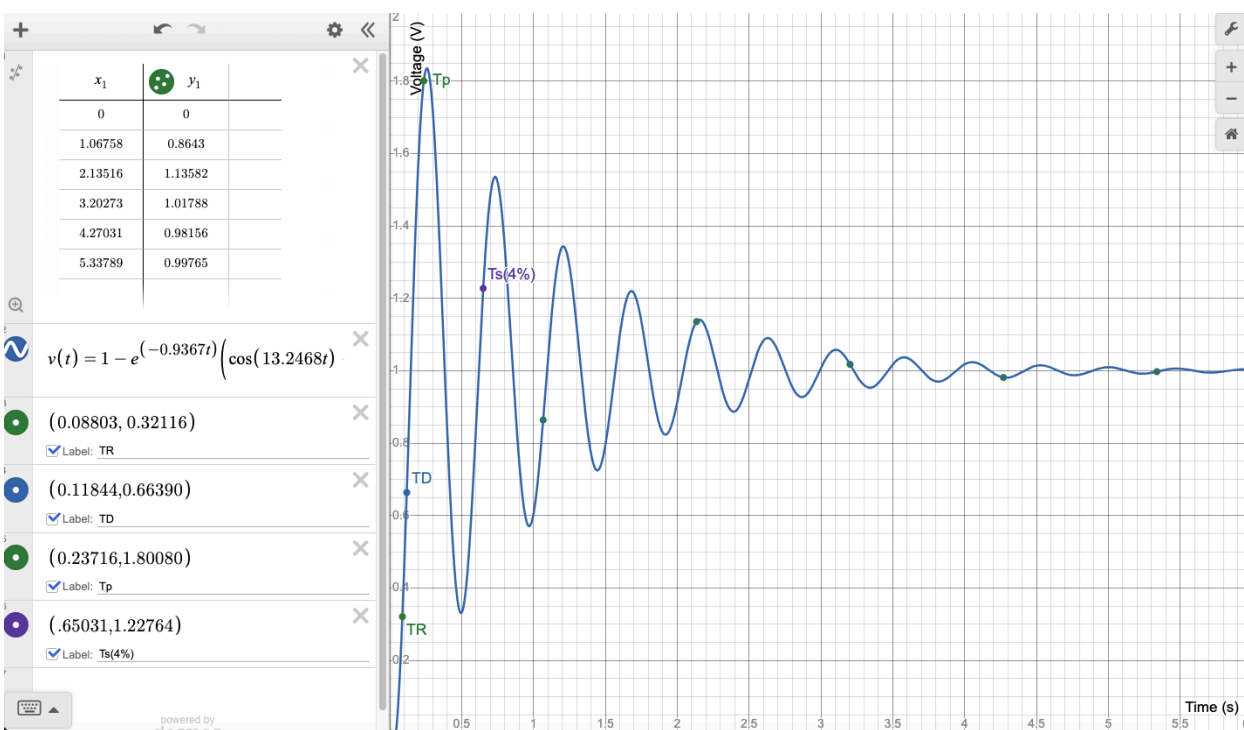
Table 13.

The step response for this underdamped system is still

$$V_o(t) = 1 - e^{-(0.94)t} (\cos 13246.8t) + \frac{0.35}{0.94} \sin(13246.8t).$$

	$V_{out}(t)$	% Final
$0\tau = 0s$	0v	0%
$1\tau = 1.0676s$	0.8643v	109.68%
$2\tau = 2.1351s$	1.1358v	114.30%
$3\tau = 3.2027s$	1.0179v	99.83%
$4\tau = 4.2703s$	0.9816v	98.05%
$5\tau = 5.3379s$	0.9977v	99.87%

Table 14. $V_{out}(t)$ after time constants



IMG 13. Please note that the step response is the same as the previous two circuits. The same underdamped, oscillatory behavior with the same aspects can be observed.

Steady state is reached after 5τ , graphed using [Desmos.com](https://www.desmos.com). Note that the equation of the graph was divided by 1000 to decrease the frequency to make the points visible with the line of best fit.

This works because reading in milliseconds stretches the graph horizontally. The equation used

in this graph is $V_o(t) = 1 - e^{-(0.94)t} (\cos 13.2468t) + \frac{0.35}{0.94} \sin(13.2468t)$ instead of

$$V_o(t) = 1 - e^{-(0.94)t} (\cos 13246.8t) + \frac{0.35}{0.94} \sin(13246.8t).$$

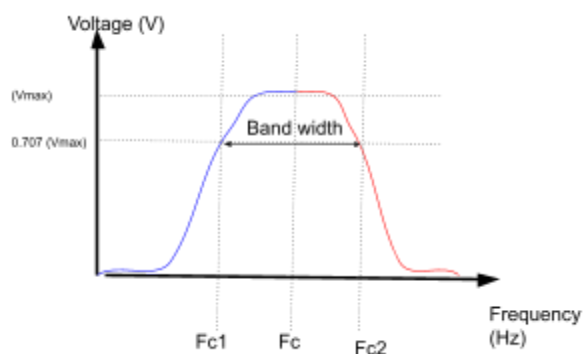
Results:

	fc1	fc2	fc	BW
Calculated	1.59 kHz	3.18 kHz	2.249 kHz	1.590 kHz
Measured	1.612kHz	15.805kHz	5.048kHz	14.193kHz

Table 15

For configuration 3, these values were obtained using the same passive component values as the previous configurations, now with the op-amp acting as a unity-gain buffer. From the Multisim Bode plot, the center frequency is inconsistent because $\sqrt{f_{c1}f_{c2}} = \sqrt{(1.612)kHz(15.805)kHz} = 5.048kHz$, and the bandwidth is $f_{c2} - f_{c1} = 15.805 - 1.612 = 14.193kHz$. The measured low cutoff frequency closely matches the calculated value, suggesting that the high pass filter portion of the circuit does behave as intended by our design. However, the measured upper cutoff frequency is much larger than expected thus influencing the measured center frequency and bandwidth to be significantly higher than the calculated values. Since this configuration uses a unity gain buffer, the discrepancy is not likely due to amplification of V_{max} and passband peak. However, it could be due to the non ideal simulated op amp behavior and aspects that impacts specifically f_{c2} . Overall, the circuit still behaves as a band pass filter but with a broader passband and unexpected simulated center frequency than the calculations predicted.

BAND PASS FILTER, FOURTH CONFIGURATION: *A second order inverting band pass filter.*



1.Determining component values

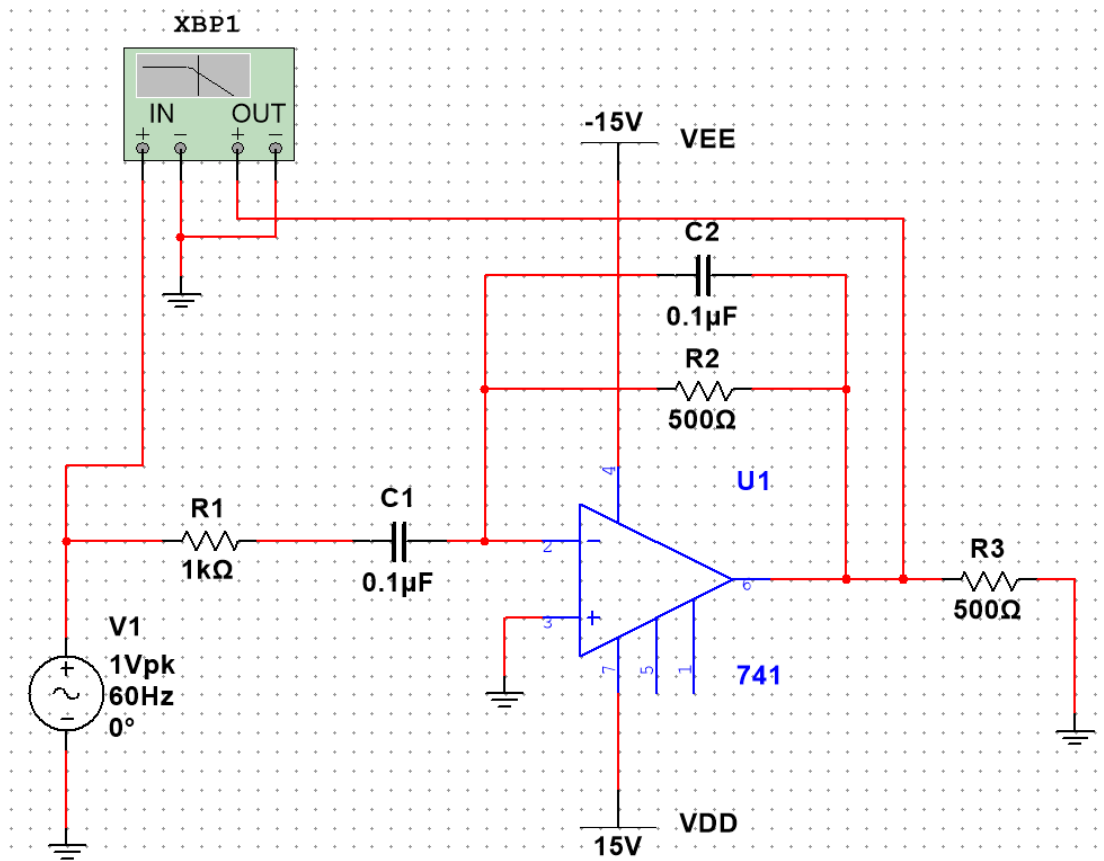
Since $F_{c1} = 1.59 \text{ kHz}$ and $F_{c2} = 3.18 \text{ kHz}$, we can keep the same component values for R1, R2, C1 and C2 as the previous configuration. Again in this configuration there is no Rf or Rin.

The gain for inverting would be calculated as $Av = \frac{-R2}{R1} = \frac{500\Omega}{1000\Omega} = -0.5$.

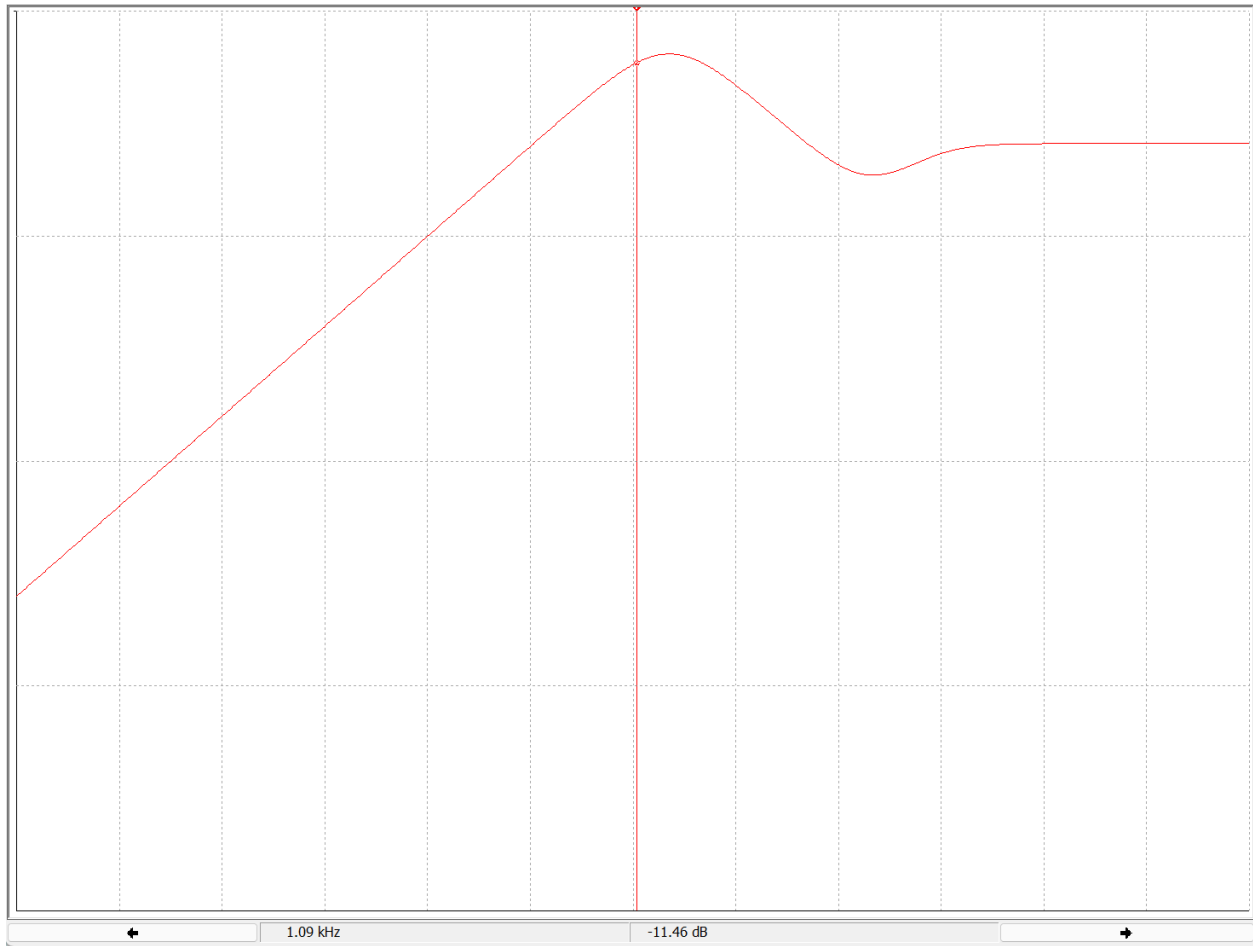
	Circuit from Multisim
C=C1=C2	0.1uF
R1	R1=2R=1000Ω
R2	R2=R=500Ω
Gain	-0.5

Table 16. Component values

Multisim



IMG 14. Schematic from Multisim

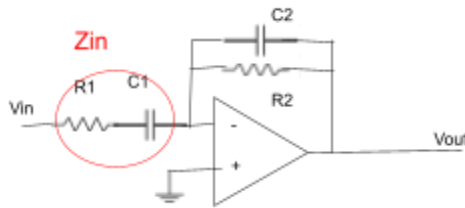
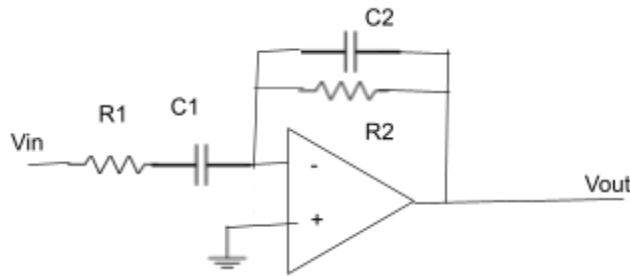


IMG 15. *Bode plot from Multisim*

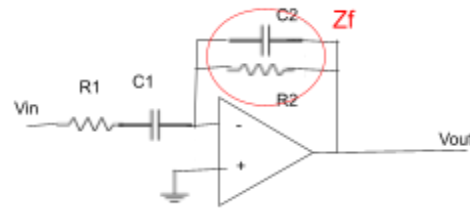
Unlike previous configurations, the amplification is not a separate stage but is “embedded”, and it attenuates the signal f_c as opposed to amplifying it. We predict that this is the cause of the dip at the center frequency.

Calculations:

First, the transfer function is determined.



$$Z_{in} = R1 + \frac{1}{C1S} = \frac{R1C1S + 1}{C1S}$$



$$Z_f = \frac{R2 \times \frac{1}{C_2 S}}{R2 + \frac{1}{C_2 S}} = \frac{R2}{R2C2S + 1}$$

$$H(s) = \frac{-Z_f}{Z_{in}} = \frac{-\frac{R2}{R2C2S + 1}}{\frac{R1C1S + 1}{C1S}} = -\frac{R2}{R2C2S + 1} * \frac{C1S}{R1C1S + 1} = -\frac{R2C1S}{(R2C2S + 1)(R1C1S + 1)}$$

Substitute $R1=2R$ and $R2=R$, and $C1=C2=C$

$$-\frac{RCS}{(RCS+1)(2RCS+1)} = -\frac{RCS}{2R^2C^2S^2 + RCS + 2RCS + 1} = -\frac{RCS}{(2R^2C^2)S^2 + (3RC)S + 1}$$

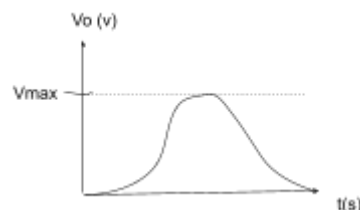
Normalize by removing the term in front of S^2 in the denominator through division

$$-\frac{RCS}{(2R^2C^2)S^2 + (3RC)S + 1} = -\frac{\frac{RCS}{2R^2C^2}}{S^2 + \frac{3RC}{2R^2C^2}S + \frac{1}{2R^2C^2}} = \frac{\frac{1}{2RC}S}{S^2 + \frac{3}{2RC}S + \frac{1}{2R^2C^2}} = H(s)$$

To check that this is indeed the transfer function of a band pass filter,

$$\text{As } S \rightarrow 0, H(s)_{total} = \frac{\frac{1}{2RC}S}{S^2 + \frac{3}{2RC}S + \frac{1}{2R^2C^2}} \approx \frac{\frac{1}{2RC}(0)}{0^2 + \frac{3}{2RC}S(0) + \frac{1}{2R^2C^2}} \approx \frac{0}{\frac{1}{2R^2C^2}} \approx 0$$

$$\text{As } S \rightarrow \infty, H(s)_{total} = \frac{\frac{1}{2RC}S}{S^2 + \frac{3}{2RC}S + \frac{1}{2R^2C^2}} \approx \frac{(\infty * \frac{1}{2RC})}{\infty^2 + (\frac{3}{2RC})(\infty) + \frac{1}{2R^2C^2}} \approx \frac{\infty}{\infty^2} \approx 0$$



Based on this analysis of the limits of s , the circuit attenuates low frequency and attenuates high frequency, thus **behaving like a band pass filter**.

$$H(s)_{total} = \frac{\frac{1}{2RC}S}{s^2 + \frac{3}{2RC}S + \frac{1}{2R^2C^2}}, \text{ so } N(Z) = (S \frac{1}{2RC}) \text{ and } N(P) = S^2 + (\frac{3}{2RC})S + \frac{1}{2R^2C^2}$$

If $N(Z) = 0$ then $0 = (S \frac{1}{2RC})$, so $S=0$ or $\frac{1}{2RC} = 0$. Therefore there is one zero at zero ($S=0$) and one zero at infinity ($\frac{1}{2RC} = 0$), indicating band pass filter behavior.

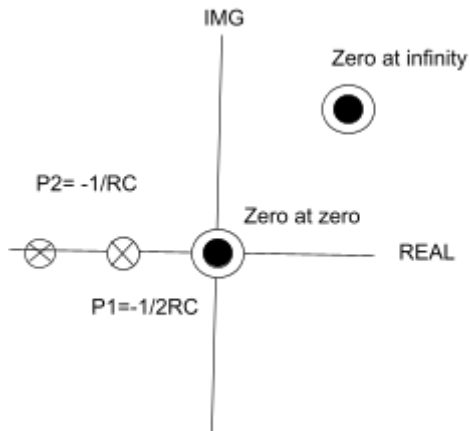
$$\text{If } N(P) = 0, 0 = S^2 + (\frac{3}{2RC})S + \frac{1}{2R^2C^2}$$

Using the quadratic formula ($\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, where $a = 1, b = \frac{3}{2RC}, c = \frac{1}{2R^2C^2}$), we get

$$\begin{aligned} \frac{-\left(\frac{3}{2RC}\right) \pm \sqrt{\left(\frac{3}{2RC}\right)^2 - 4(1)\left(\frac{1}{2R^2C^2}\right)}}{2(1)} &= \frac{-\left(\frac{3}{2RC}\right) \pm \sqrt{\frac{9}{4R^2C^2} - \frac{2}{R^2C^2}}}{2} = \frac{-\left(\frac{3}{2RC}\right) \pm \sqrt{\frac{9}{4R^2C^2} - \frac{8}{4R^2C^2}}}{2} = \frac{-\left(\frac{3}{2RC}\right) \pm \sqrt{\frac{1}{4R^2C^2}}}{2} \\ &= \frac{-\left(\frac{3}{2RC}\right) \pm \frac{1}{\sqrt{4R^2C^2}}}{2} = \frac{-\left(\frac{3}{2RC}\right) \pm \frac{1}{2RC}}{2} \end{aligned}$$

$$\text{Pole 1: } S1 = \frac{-\left(\frac{3}{2RC}\right) + \frac{1}{2RC}}{2} = \frac{-1}{2RC}$$

$$\text{Pole 2: } S2 = \frac{-\left(\frac{3}{2RC}\right) - \frac{1}{2RC}}{2} = \frac{-1}{RC}$$



To determine if the system is underdamped, critically damped, or over damped we can calculate the quality factor (Q)

$$Q = \frac{fc}{BW} = \frac{fc}{fc_2 - fc_1} = \frac{\sqrt{f_2 f_1}}{f_2 - f_1} = \frac{\sqrt{(1.59k)(3.18k)}}{(3.18k - 1.59k)} = \frac{2248.6}{1590} = 1.41$$

$Q > \frac{1}{2}$	Stable and underdamped
$Q = \frac{1}{2}$	Stable and critically damped
$Q < \frac{1}{2}$	Stable and overdamped

Therefore, the system is stable and underdamped. To check, we can find the dampening ratio

$$\xi = \frac{1}{2Q} = \frac{1}{2.83} = 0.35$$

Underdamped	Critically damped	Overdamped
$0 < \xi < 1$	$\xi = 1$	$\xi > 1$

The dampening ratio confirms that the system is underdamped. As just the gain changed, we still have the same underdamped characteristics and same step response compared to the configurations in the first three circuits. Therefore, the mathematical work is identical, and so the results sans process is shown below. **For the calculation process, please refer to the calculation process of the first circuit configuration on page 19.**

$$\omega_1 = 10,000 \text{ rad/sec and } \omega_2 = 20,000 \text{ rad/sec, so } \omega_n = 14142.14 \text{ rad/sec}$$

	Formula	Result
Dampening factor	$\beta = \sqrt{1 - \xi^2}$	$\beta = 0.9367$
	$\omega_d = \beta \omega_n$	$\omega_d = 13247 \text{ rad/s}$
Rise time	$T_R = \frac{1 + 0.7\xi}{\omega_n}$	$T_R = 0.088 \text{ ms}$
Delay time	$T_D = \frac{0.8 + 2.5\xi}{\omega_n}$	$T_D = 0.1184 \text{ ms}$
Peak time	$T_p = \frac{\pi}{\omega_d}$	$T_p = 0.237 \text{ ms}$

% Overshoot	$O.S\% = e^{\frac{-\pi\xi}{\beta}} * 100$	$O.S\% = 30.9\%$
Settling time	$t_s = \frac{-\ln(\% \text{ in decimal})}{\xi\omega_n}$	2%: $t_s = 0.790 \text{ ms}$
		4%: $t_s = 0.650 \text{ ms}$
		5%: $t_s = 0.605 \text{ ms}$

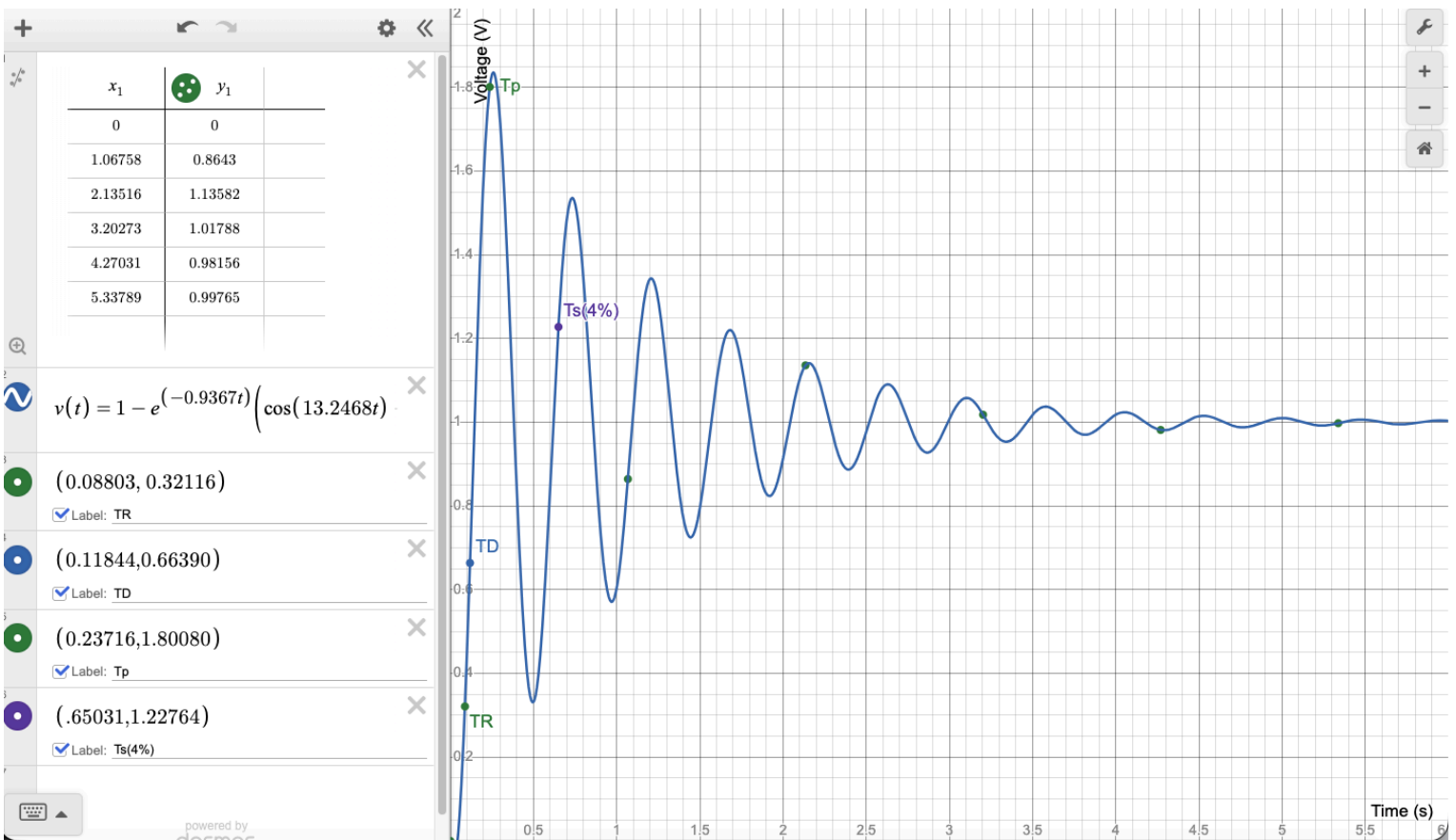
Table 17.

The step response for this underdamped system is still

$$V_o(t) = 1 - e^{-(0.94)t} (\cos 13246.8t) + \frac{0.35}{0.94} \sin(13246.8t).$$

	$V_{out}(t)$	% Final
$0\tau = 0s$	0v	0%
$1\tau = 1.0676s$	0.8643vv	109.68%
$2\tau = 2.1351s$	1.1358v	114.30%
$3\tau = 3.2027s$	1.0179v	99.83%
$4\tau = 4.2703s$	0.9816v	98.05%
$5\tau = 5.3379s$	0.9977v	99.87%

Table 18. $V_{out}(t)$ after time constants



IMG 17. Please note that the step response is the same as the previous three circuits. The same underdamped, oscillatory behavior with the same aspects can be observed.

Steady state is reached after 5τ , graphed using [Desmos.com](https://www.desmos.com). Note that the equation of the graph was divided by 1000 to decrease the frequency to make the points visible with the line of best fit.

This works because reading in milliseconds stretches the graph horizontally. The equation used

in this graph is $V_o(t) = 1 - e^{-(0.94)t} (\cos 13.2468t) + \frac{0.35}{0.94} \sin(13.2468t)$ instead of

$$V_o(t) = 1 - e^{-(0.94)t} (\cos 13246.8t) + \frac{0.35}{0.94} \sin(13246.8t).$$

Results:

The Bode plot for the fourth configuration still shows band pass behavior. Additionally, the calculated results and analysis of circuit behavior and transfer function also prove band pass behavior. Unlike the previous configurations, the Bode plot (page 42) shows a dip near the center frequency instead of a v_{max} peak amplitude. Our guess about why this is happening is because the gain is embedded in the inverting filter structure and is -0.5, meaning that it attenuates rather than amplifies the signal in the passband due to being less than a magnitude of 1. This helps explain some of the discrepancy shown in the Multisim plot. There is not a clear bandwidth in the Bode plot, unfortunately. This is due to the lack of a clear upper and lower frequency, as the bandwidth is calculated as $\sqrt{f_{c1} * f_{c2}}$. The center frequency seems to be roughly 1.09kHz, as opposed to our calculated 2.249kHz.

Conclusion

Throughout this lab our primary objective was to design and analyze multiple second order active band pass filter configurations. Although we had previous experience with understanding how component selection influences circuit behavior through choosing the cutoff frequency and gain, this lab introduced us to component selection consequences specifically on bandwidth. This was a struggle for us, as our calculations did not seem to match the Bode plot outputs, and we struggled to identify why and what influences were present on the passband and other circuit design aspects. This taught us that proper filter design requires consideration beyond simply selecting values for the center frequency, and that simulation software Bode plot results cannot be trusted to match the calculation process.

For Configuration 1, the measured and calculated center frequencies closely matched confirming that the selected component values were appropriate and that the circuit *generally* functioned as intended. However, the measured lower and upper cutoff frequencies, as well as the bandwidth, showed significant deviation from the theoretical values. We believe that a possible explanation for this discrepancy is that the total gain of both amplifier stages combined produced an overall gain of 4, which increased V_{max} (located where the center frequency is) and may have shifted the -3 dB cutoff points resulting in a wider measured bandwidth. Similarly, Configuration 2 produced a measured center frequency close to the calculated center frequency, again suggesting that the circuit operated near the intended passband center. However, once again the measured passband was significantly broader than predicted. We hypothesize that the gain of 2 (from stage 2) once again increased V_{max} (located where the center frequency is) and may have shifted the -3 dB cutoff points resulting in a wider measured bandwidth.

For Configuration 3, the lower cutoff frequency closely matched the calculated value indicating that the high-pass portion of the filter behaved as intended. However, the upper cutoff frequency was significantly larger than expected which unfortunately caused the measured center frequency and bandwidth to be much greater than predicted by calculations. Despite this discrepancy from the Multisim Bode plots, the circuit still exhibited band-pass behavior with a broader-than-expected passband. For Configuration 4, the Bode plot still demonstrated band-pass behavior, which was further confirmed by the calculated transfer function and circuit analysis. However, unlike the previous configurations, the Bode plot showed a dip near the center frequency instead of a clear maximum amplitude peak. We theorize that this is because the gain in this configuration is embedded in the inverting filter structure, and is equal to -0.5 . Since the

magnitude of this gain is less than 1, we think that the circuit attenuates rather than amplifies the center frequency, and that this attenuation likely contributed to the unusual appearance of the Bode plot. However, there are likely other factors we aren't aware of that are causing more discrepancies in the Multisim results.

Overall, the calculations throughout the lab changed minimally between configurations because the passive component values remained identical due to the continued usage of the lower and upper cutoff frequencies. Pertaining to the calculations the poles and zeros remained the same, as well as the calculation analysis of the circuit behavior, the dampening ratio, Q , underdamped aspects, and steady state graph. However, changing the amplifier and the configuration between non-inverting, inverting, and unity-gain buffer arrangements did alter the transfer function and amplification, and resulted in differing success in Multisim output. Ultimately, this lab showed us that although transfer function calculations may remain similar between designs, real, calculated and simulated circuit behavior can vary significantly depending on gain, op-amp configuration, and impacts causing non-ideal implementation.

References:

- **Circuit Analysis in the Frequency Domain & Analog Filter Design**

M. Mousavi, *Circuit Analysis in Frequency Domain & Analog Filter Design*, The Electromechanical System Engineering Technology Department, California State Polytechnic University, Pomona.

- Daycounter, Inc. “Sallen-Key High Pass Filter Design Equations.” Daycounter, <https://www.daycounter.com/Filters/SallenKeyHP/Sallen-Key-HP-Filter-Design-Equations.phtml>. Accessed 12 Apr. 2026.
- Daycounter, Inc. “Sallen-Key Low Pass Filter Design Equations.” Daycounter, <https://www.daycounter.com/Filters/SallenKeyLP/Sallen-Key-LP-Filter-Design-Equations.phtml>. Accessed 12 Apr. 2026.
- Jotrin Electronics, “Op-Amp LM358P: Pinout, Circuits, and Datasheet.” <https://www.jotrin.com/technology/details/op-amp-lm358p-pinout-circuits-datasheet> Accessed: March 8, 2026.
- Desmos Studio PBC, “Desmos Graphing Calculator.” <https://www.desmos.com/calculator> Accessed: March 8, 2026.